

Fish CDK2 recruits Dtx4 to degrade TBK1 through ubiquitination in the antiviral response

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Long-Feng Lu, Can Zhang, Zhuo-Cong Li, Bao-Jie Cui, Yang-Yang Wang, Ke-Jia Han, Xiao Xu, Chu-Jing Zhou, Xiao-Yu Zhou, Yue Wu, Na Xu, Xiao-Li Yang, Dan-Dan Chen, Xi-Yin Li, Li Zhou, Shun Li 

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China • University of Chinese Academy of Sciences, Beijing, China • College of Fisheries and Life Science, Dalian Ocean University, Dalian, China • State Key Laboratory of Freshwater Ecology and Biotechnology, Institute of Hydrobiology, Wuhan, China • Key Laboratory of Aquaculture Disease Control, Ministry of Agriculture, Wuhan, China • Laboratory for Marine Biology and Biotechnology, Qingdao Marine Science and Technology Center, Qingdao, China

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eLife Assessment

This **valuable** study uses zebrafish as a model to reveal a role for the cell cycle protein kinase CDK2 as a negative regulator of type I interferon signaling. The evidence supporting the authors' claims is **convincing**, including both in vivo and in vitro investigative approaches that corroborate a role for CDK2 in regulating TBK1 degradation. In this latest version, the authors included data addressing concerns raised by the reviewers at the first peer review round that strengthens the conclusions. This work will interest cell biologists, immunologists, and virologists.

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Abstract

Although the classical biological protein cell cycle protein kinase CDK2 has been extensively studied in higher vertebrates, its function in lower vertebrates beyond the regulation of mitosis remains unknown. In this study, we report a distinct mechanism whereby IFN expression is negatively regulated in fish by CDK2. After infection with the spring viremia of carp virus (SVCV), fish CDK2 expression significantly increased in tissues and cells. Moreover, antiviral resistance was improved in *cdk2*^{-/-} homozygotes, and the antiviral cytokine interferon (IFN) expression was significantly higher. At the cellular level, CDK2 overexpression reduced IFN expression, while *cdk2* knockdown increased the ability of cells to produce IFN. Subsequently, it was discovered that fish CDK2 binds and degrades TBK1, resulting in reduced IFN. CDK2 increases the K48-linked ubiquitination of TBK1, causing its degradation, while E3 ubiquitin ligase Dtx4 was found to be involved in this process following the significant enhancement of TBK1 K48-linked ubiquitination. Protein mass spectrometry and immunoblot analysis confirmed that the K567 site on TBK1 is essential for CDK2 to engage with Dtx4 and degrade TBK1; thus, after mutating the K567 site, K48-linked ubiquitination of TBK1 was not enhanced by Dtx4, and TBK1 was not degraded by CDK2. Our

data demonstrate that fish CDK2 recruits the E3 ubiquitin ligase Dtx4 to target the K567 site of TBK1 and promote its degradation. These results suggest that CDK2 in lower vertebrates is implicated in a specialized role for antiviral innate immunity.

Introduction

Interferon (IFN) production is central to the host's innate immune response to viral infections. IFN acts as a ligand, binding to receptors on neighboring cell membranes through the autocrine and paracrine pathways, which initiates the Janus kinase/signal transducer and activator of the transcription (JAK/STAT) signaling pathway¹. This, in turn, leads to the transcription of a large number of antiviral genes, ultimately resulting in the clearance of intracellular viral components². IFN induction depends on cellular pattern recognition receptors (PRRs) and sensing of conserved pathogen-associated molecular patterns (PAMPs), with the retinoic acid-inducible gene I (RIG-I)-like receptors (RLRs) signaling pathway being critical in this system³. Viral nucleic acids are recognized by the RLRs, which then initiate a series of signaling events that lead to downstream IFN expression. TANK-binding kinase 1 (TBK1) efficiently phosphorylates IFN regulatory factor 3/7 (IRF3/7), facilitating its nuclear entry⁴.

Effective control of IFN production is crucial during viral infection in maintaining immune responses and homeostasis, whereby excessive IFN can cause inflammatory storms that harm the organism. Hence, TBK1 plays a significant role in the IFN signaling pathway and is negatively regulated by various factors to balance IFN expression in the host⁵. Mammalian TBK1 is degraded by NLRP4, which recruits the E3 ubiquitin ligase DTX4 for K48-linked polyubiquitination at Lys670⁶. Additionally, DYRK2 is crucial in the degradation of TBK1 by NLRP4 through the phosphorylation of Ser527, as mentioned in previous studies^{7,8}. The function of TBK1 is conserved in fish and subject to modulation by multiple molecules to prevent excessive IFN expression. Indeed, TMEM33 in zebrafish serves as a TBK1 substrate, reducing IRF3 phosphorylation and hindering TBK1 kinase activity to diminish IFN expression⁹. Generally, the ability of TBK1 to induce IFN is conserved and tightly regulated within the host.

CDKs are a family of Ser/Thr kinases that act as cell cycle regulators. Cell cycle progression from the G1 phase to the S phase and from G2 to mitosis is controlled by various cyclin-CDK complexes. For example, CDK2 and cyclin E regulate the G1-S transition¹⁰. Further studies have unveiled other functions of the CDK family in biological processes beyond cell cycle regulation. The transcription of multiple proinflammatory genes is upregulated in a CDK-dependent fashion throughout the G1 phase¹¹. In the context of antitumor immunity in fibrosarcoma and lung carcinoma, inhibiting CDK2 leads to the RB protein being phosphorylated, which results in increased production of type I IFN¹². Another study demonstrated that inhibiting CDKs or the knockdown of CDKs activities caused a significant blockade in IFN release from culture supernatants¹³. While there are several reported functions of CDKs in innate immunity, understanding the roles of CDKs in this area remains unclear.

The immune systems of higher vertebrates (e.g., humans) and lower vertebrates (e.g., fish) generally exhibit some consistency, although there are notable differences. For instance, IFN phosphorylates fish IRF3, which is only phosphorylated by viruses in mammals. Additionally, fish MVP functions as an IFN-negative regulator, while human MVP mediates IFN-positive expression^{14,15,16,17}. There are 13 CDK genes in the human genome and 21 CDK orthologs in zebrafish, indicating that CDK functions are conserved and differ between fish and humans. Our report reveals that fish CDK2 acts as an IFN negative regulator that recruits Dtx4 to facilitate the ubiquitination of TBK1, thereby restricting IFN expression. These findings offer valuable insights into the various roles of CDK2.

Results

1. CDK2 is upregulated during viral infection *in vivo* and *in vitro*

To identify potential molecules linked with viral infection, we generated a transcriptome pool from liver and spleen tissues taken from SVCV-infected zebrafish. Notably, classical cell cycle regulator *cdk2* was among the few upregulated genes in the CDK family (Fig. 1A and B). In previous studies, it was found that mammalian *cdk2* was not regulated during viral infection. As shown in Figure 1C, this was confirmed by detecting *cdk2* after THP-1 cells infection with VSV, and it was observed that *cdk2* was not upregulated. Fish *cdk2* was significantly increased upon infection with fish viruses CyHV-2 or SVCV in *Carassius auratus gibelio*, *Danio rerio*, and *Pimephales promelas*, compared to human *cdk2*, indicating that fish CDK2 is involved in the host's antiviral response. Meanwhile, specific qPCR and immunoblot analysis to validate this finding in zebrafish tissues, which demonstrated that the *cdk2* transcript level was significantly increased upon SVCV infection. The CDK2 protein level was also consistently higher in the viral infection group (Fig. 1D and 1E). We investigated whether this CDK2 expression pattern also existed *in vitro*. For this purpose, we used the zebrafish embryonic fibroblast cell line (ZF4) and another cyprinid fish cell line (epithelioma papulosum cyprini (EPC)). During the 48-hour (h) viral infection period, CDK2 was significantly upregulated at the protein level (Fig. 1F). Taken together, the *in vivo* and *in vitro* data suggest that fish CDK2 may play a role in the response to viral infection.

2. *cdk2*^{-/-} fish exhibit effective antiviral capacities

To investigate the role of CDK2 in the antiviral process, we generated a *cdk2* knockout zebrafish homozygote (*cdk2*^{-/-}). Our findings demonstrated that the survival rate under viral infection was significantly higher in the *cdk2*^{-/-} group than in the wild-type group (Fig. 2A). We selected the liver, spleen, and kidney as representative tissues in which to analyze the *cdk2*^{-/-} antiviral capacity. In the H&E staining assay, severe tissue damage was observed in the wild-type group, while the *cdk2*^{-/-} group displayed dramatically less tissue damage (Fig. 2B). The viral transcripts in these tissues were observed and then compared to the abundance in the wild-type. The replication of viral genes, such as SVCV N, was typically inhibited in *cdk2*^{-/-} homozygote tissues (Fig. 2C). Viral protein level analysis also confirmed these results, as SVCV G, N, and P proteins were suppressed in *cdk2*^{-/-} homozygotes (Fig. 2D).

Transcriptomics analysis was employed to investigate the antiviral regulation by CDK2. Differential expression analysis identified 1707 upregulated genes and 1490 downregulated genes in the liver of *cdk2* knockout zebrafish versus WT zebrafish after SVCV infection (Fig. 2E). Moreover, gene set enrichment analyses (GSEAs) were performed to analyze the IFN response to virus infection-related genes, which demonstrated that these genes were significantly activated in the *cdk2*^{-/-} group (Fig. 2F). Differential expression analysis further showed that *cdk2* knockout results in the upregulation of many IFN-stimulated genes (ISGs) after SVCV infection (Fig. 2G). To procure the transcriptome data, IFN was assayed in zebrafish tissues, including liver, spleen, and kidney. Similar to in the transcriptome assay, IFN was found to be significantly higher in the *cdk2*^{-/-} groups, indicating that stronger antiviral capacity and higher IFN transcription occurred in *cdk2*^{-/-} fish (Fig. 2H). Collectively, these data suggest that viral replication was hindered in the *cdk2*^{-/-} fish.

3. CDK2 inhibits IFN expression and promotes viral proliferation

The impact of CDK2 on viral infection was examined in relation to IFN. The IFN response is an essential mechanism for host resistance against viruses. Overexpression of CDK2 decreased the IFN promoter and ISRE motif activation by SVCV or poly I:C (Fig. 3A). An effective shcdk2 was

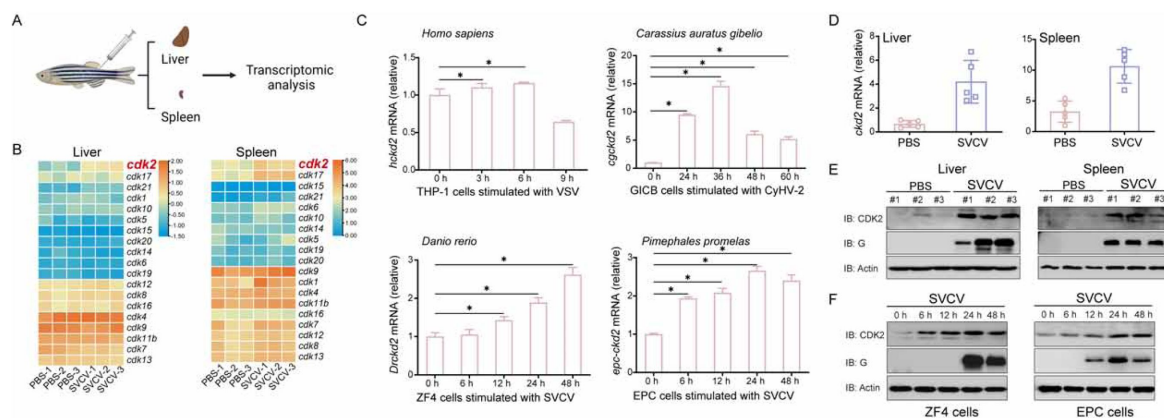


Figure 1.

In vivo and in vitro, CDK2 is upregulated during viral infection.

(A) Schematic representation of zebrafish tissue dissection and RNA extraction for transcriptome sequencing. The liver and spleen tissues from male and female zebrafish injected with PBS (5 μ L/individual), SVCV (5×10^8 TCID₅₀/mL, 5 μ L/individual) for 48 h. Total RNAs were extracted and used for transcriptome sequencing and analysis. (B) Heatmap view of mRNA variations of CDKs in the liver and spleen of zebrafish with or without SVCV infection. (C) qPCR analysis of *cdk2* mRNA in the THP-1, GICB, ZF4, or EPC cells infected with VSV, CyHV-2, or SVCV for the indicated times. (D) qPCR analysis of *cdk2* mRNA in the liver and spleen of zebrafish (n = 5 per group) given injected intraperitoneally (i.p.) with PBS or SVCV for 48 h. (E) IB of proteins in the liver and spleen of zebrafish (n = 3 per group) given i.p. injection of PBS or SVCV for 48 h. (F) IB of proteins in ZF4 and EPC cells infected with SVCV for the indicated times.

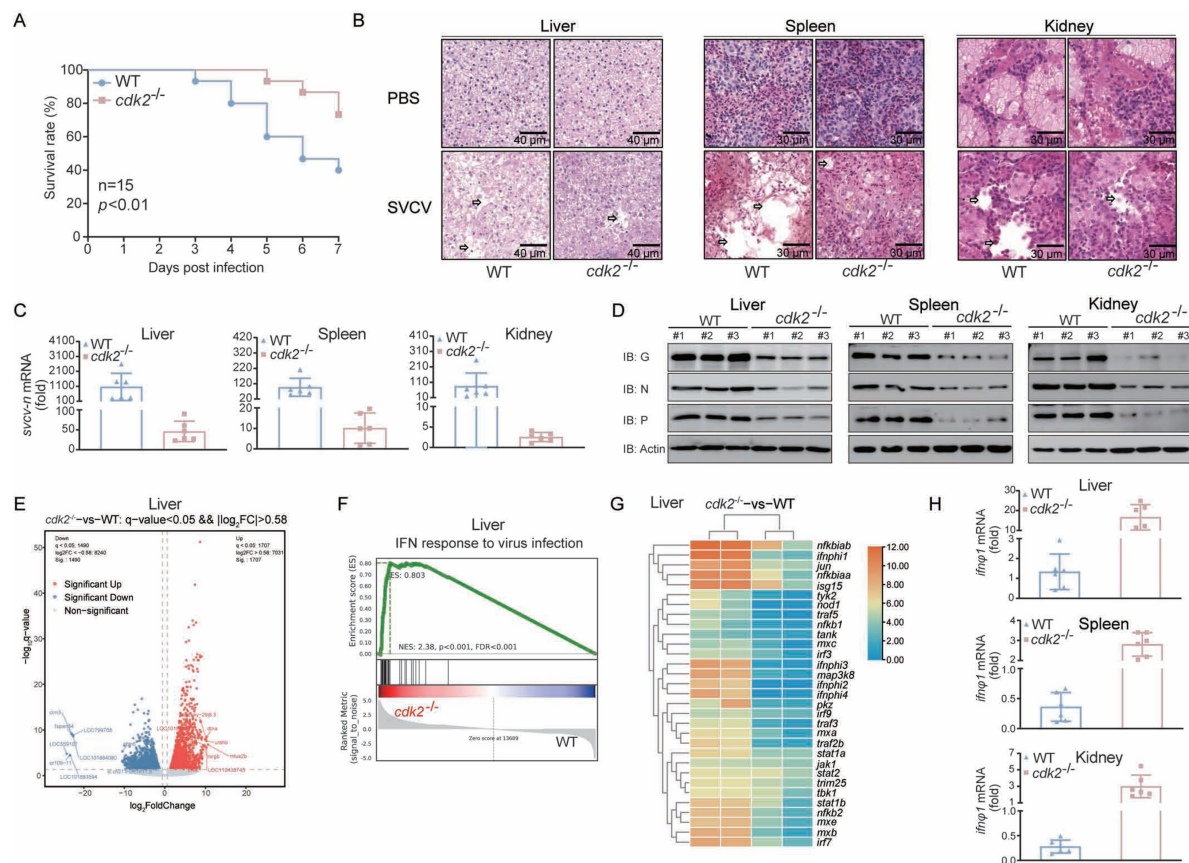


Figure 2.

CDK2 deficiency protects fish from SVCV infection.

(A) Survival (Kaplan-Meier Curve) of *cdk2*^{+/-} and *cdk2*^{-/-} zebrafish (n = 15 per group) at various days after i.p. infected with SVCV (5×10^8 TCID₅₀/mL, 5 μL/individual). (B) Microscopy of H&E-stained liver, spleen, and kidney sections from *cdk2*^{+/-} and *cdk2*^{-/-} zebrafish treated with SVCV for 48 h. (C) qPCR analysis of *svcv-n* mRNA in the liver, spleen, and kidney of *cdk2*^{+/-} and *cdk2*^{-/-} zebrafish (n = 6 per group) given i.p. injection of SVCV for 48 h. (D) IB of proteins in the liver, spleen, and kidney sections from *cdk2*^{+/-} and *cdk2*^{-/-} zebrafish (n = 3 per group) treated with SVCV for 48 h. (E) CDK2 regulates antiviral response-relevant target genes, presented as a volcano plot of genes with differential expression after SVCV infection in the liver of *cdk2*^{+/-} and *cdk2*^{-/-} zebrafish. (F) GSEA of differentially expressed genes in the liver of *cdk2*^{+/-} and *cdk2*^{-/-} zebrafish with SVCV infection and enrichment of IFN. FDR (q-value) was shown. (G) Heatmap view of mRNA variations of IFN-mediated ISG sets in the liver of *cdk2*^{+/-} and *cdk2*^{-/-} zebrafish with SVCV infection. (H) qPCR analysis of *ifnφ1* mRNA in the liver, spleen, and kidney of *cdk2*^{+/-} and *cdk2*^{-/-} zebrafish (n = 6 per group) given i.p. injection of SVCV for 48 h.

produced and identified (Fig. 3B). The impact of CDK2 knockdown was explored, with the results suggesting that the downregulation of CDK2 facilitates IFN promoter activity (Fig. 3C). The mRNA levels of *ifn* and downstream ISG *vig1* transcription were monitored, revealing that CDK2 caused a significant decrease in *ifn* and *vig1*, whereas the knockdown of CDK2 increased the IFN response (Fig. 3D). In the antiviral capacity assays, CDK2 from zebrafish, gibel carp, and grass carp all promoted the proliferation of their respective corresponding viruses (Fig. 3E). Conversely, CDK2 knockdown significantly suppressed viral proliferation (Fig. 3F). Cells overexpressing CDK2 showed enhanced production of viral mRNA and proteins, while CDK2-knockdown cells showed attenuation of the same viral mRNA and proteins (Fig. 3G and H). Immunofluorescence revealed a higher intensity of green signals indicating SVCV N protein in the CDK2 overexpression group compared to the control, while a lower green signal was observed in the CDK2-knockdown group compared to the normal group (Fig. 3I). These findings suggest that CDK2 inhibits IFN expression in the host and reduces antiviral capacity.

4. CDK2 interacts with TBK1 and reduces its expression

The RLR pathway plays an important role in IFN activation. To investigate whether CDK2 counters RLR signaling, as observed earlier, we examined whether CDK2 inhibits RLR factors that activate IFN promoters. The activation of the IFN promoters induced by MAVS and TBK1 was significantly inhibited by CDK2, whereas it was unaffected by MITA (Fig. 4A). CDK2 knockdown restored MAVS and TBK1 functions following IFN promoter stimulation (Fig. 4B). Moreover, the *ifn* and *vig1* transcripts were monitored. The results showed that CDK2 notably reduced *ifn* and *vig1* mRNA levels, whereas the CDK2 knockdown improved the IFN response (Fig. 4C and D). These findings suggest that CDK2 targets either MAVS or TBK1. The subcellular localization of CDK2 was observed to be distributed throughout the cell. However, TBK1 and MAVS exhibited cytoplasmic localization, while there appeared to be a significant punctate overlap between CDK2 and TBK1, indicating a possible association between CDK2 and TBK1 (Fig. 4E). Subsequently, association analysis of CDK2 and MAVS or TBK1 was conducted. The co-IP assay revealed an interaction between CDK2 and TBK1, yet not between CDK2 and MAVS (Fig. 4F). Afterward, an endogenous co-IP assay was performed to confirm the interaction between CDK2 and TBK1, while the interaction was also verified with TBK1 or CDK2 being enriched (Fig. 4G). To identify the essential domain that mediates the interaction with CDK2, two truncated TBK1 mutants were generated (Fig. 4H). The TBK1-ΔN-flag, which lacked the kinase domain, meaning it cannot bind to CDK2, indicated that the kinase domain in TBK1 is necessary for it to associate with CDK2 (Fig. 4I). Since the interaction between CDK2 and TBK1 was confirmed, whether CDK2 affects TBK1 stability was also investigated. Co-overexpression of the relevant RLR factors and CDK2 demonstrated that TBK1 expression was substantially decreased in the presence of CDK2, compared to MAVS (Fig. 4J). Consistently, CDK2 impaired endogenous TBK1 under both normal and stimulation states, whereas CDK2 knockdown abolished this effect, indicating that CDK2 impaired TBK1 expression (Fig. 4K and L). The expression of TBK1 was also significantly increased in the liver, spleen, and kidney of *cdk2*^{-/-} fish, confirming the *in vivo* impact of CDK2 on TBK1 expression (Fig. 4M). Subsequently, its biological function was studied to investigate whether CDK2 affects the crucial antiviral role of TBK1. During SVCV infection, cells overexpressing TBK1 showed little CPE; however, CDK2 dramatically counteracted the antiviral capacity of TBK1, as confirmed by virus titer identification (Fig. 4N). For viral protein expression, CDK2 restored the TBK1-mediated prevention of viral protein expression, including SVCV G, N, and P proteins. Immunofluorescence also demonstrated that the inhibition of SVCV N protein by TBK1 was restored following CDK2 overexpression (Fig. 4O and P). CDK2 also restored viral nucleic acid transcription, which was abolished by TBK1 (Fig. 4Q). These results demonstrate that CDK2 interacts with the kinase domain in TBK1, leading to reduced expression and a weakened antiviral effect.

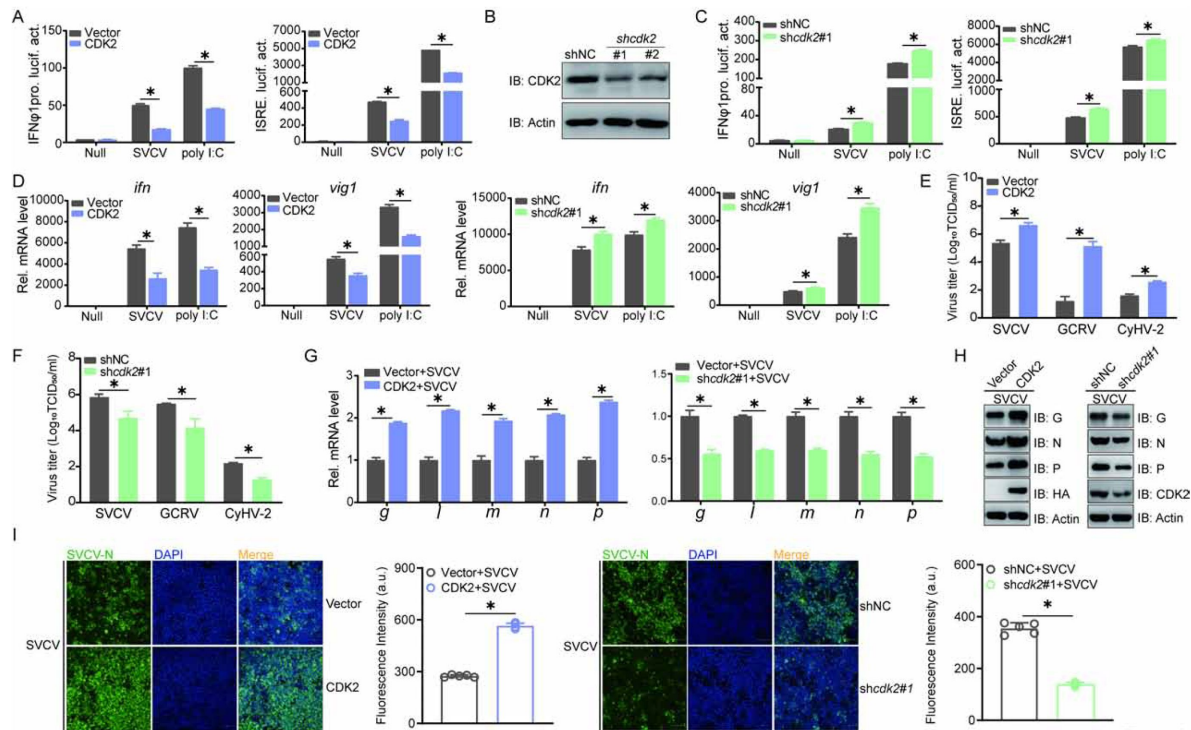


Figure 3.

CDK2 negatively regulates IFN production and promotes viral replication.

(A and C) Luciferase activity of IFN ϕ 1pro and ISRE in EPC cells transfected with indicated plasmids for 24 h, and then untreated or infected with SVCV (MOI = 1) or transfected with poly I:C (0.5 μ g) for 24 h before luciferase assays. (B) IB of proteins in EPC cells transfected with indicated plasmids for 24 h. (D) qPCR analysis of *ifn* and *vig1* in EPC cells transfected with indicated plasmids for 24 h, and then untreated or infected with SVCV (MOI = 1) or transfected with poly I:C (2 μ g) for 24 h. (E and F) Plaque assay of virus titers in EPC, CIK, and GiCB cells transfected with indicated plasmids for 24 h, followed by SVCV, GCRV, and CyHV-2 challenge for 24–72 h. (G and H) qPCR and IB analysis of SVCV genes in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. (I) IF analysis of N protein in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. The fluorescence intensity (arbitrary unit, a.u.) was recorded by the LAS X software, and the data were expressed as mean $\pm$ SD, n = 5.

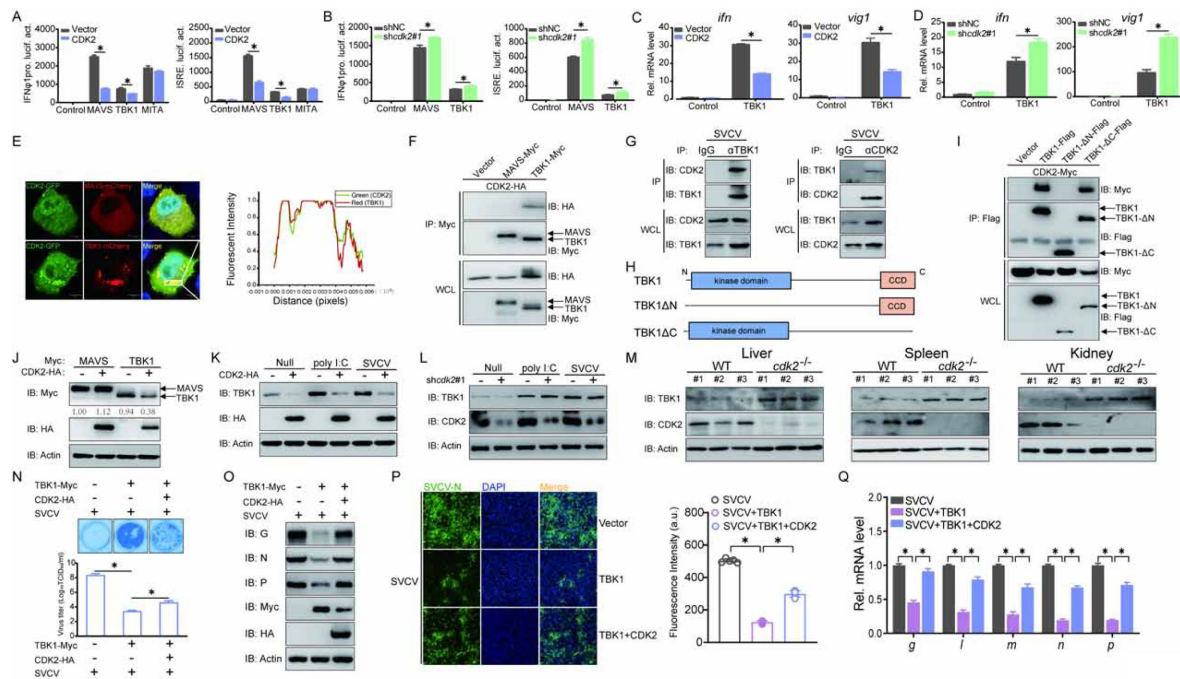


Figure 4.

CDK2 associates with TBK1 and mediates its degradation.

(A and B) Luciferase activity of IFN ϕ 1pro and ISRE in EPC cells transfected with indicated plasmids for 24 h. (C and D) qPCR analysis of *ifn* and *vig1* in EPC cells transfected with indicated plasmids for 24 h. (E) Confocal microscopy of CDK2 and TBK1 in EPC cells transfected with indicated plasmids for 24 h. The coefficient of colocalization was determined by qualitative analysis of the fluorescence intensity of the selected area in Merge. (F) IB of WCLs and proteins immunoprecipitated with anti-Myc antibody-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (G) IB of WCLs and proteins immunoprecipitated with anti-TBK1 or anti-CDK2 antibody from EPC cells infected with SVCV for 24 h. (H) Schematic representation of full-length TBK1 and its mutants. (I) IB of WCLs and proteins immunoprecipitated with anti-Flag antibody-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (J) IB of proteins in EPC cells transfected with indicated plasmids for 24 h. (K and L) IB of proteins in EPC cells transfected with CDK2-HA or *shcdk2#1* for 24 h, followed by untreated or infected with SVCV (MOI = 1) or transfected with poly I:C (2 μ g) for 24 h. (M) IB of proteins in the liver, spleen, and kidney sections from *cdk2*^{+/+} and *cdk2*^{-/-} zebrafish (n = 3 per group). (N) Plaque assay of virus titers in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24-48 h. (O and Q) IB and qPCR analysis of SVCV genes in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. (P) IF analysis of N protein in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. The fluorescence intensity (arbitrary unit, a.u.) was recorded by the LAS X software, and the data were expressed as mean $\pm$ SD, n = 5.

5. CDK2 causes TBK1 degradation by increasing K48-linked polyubiquitination

The next step was to investigate the mechanisms through which CDK2 negatively regulates TBK1. To determine whether the decrease in TBK1 occurred at the mRNA or protein level, *tbk1* transcription was monitored. CDK2 had little effect on *tbk1* in either the control or virus-infected groups (**Fig. 5A**). Therefore, attention was placed on modulating TBK1 at the protein level. Various reagents that inhibit the ubiquitin (Ub)-proteasome and autophagosome, such as MG132, 3-MA, Baf-A1, and CQ, were utilized to clarify the precise regulatory mechanism. Compared to the autophagosome inhibitors 3-MA, Baf-A1, and CQ, using the Ub-proteasome inhibitor MG132 significantly impeded TBK1 degradation in a dose-dependent manner (**Fig. 5B** and **C**). This suggests that the degradation of TBK1 by CDK2 is proteasome-dependent. Similar results were observed for endogenous TBK1 in the presence or absence of poly I:C stimulation or SVCV infection (**Fig. 5D**). Since ubiquitination is an important process during proteasome-dependent degradation, we next investigated whether ubiquitination was important in the CDK2-mediated degradation of TBK1. Expectedly, immunoblot analysis confirmed that CDK2 increased the ubiquitination of TBK1 (**Fig. 5E**). Polyubiquitin chain modification, either K48-linked or K63-linked, can either target proteins for degradation or increase stability. To investigate how TBK1 was modified, we performed MS analysis of ubiquitinated TBK1 from cells and found that K48-linked polyubiquitinated TBK1 was readily detected in CDK2-overexpressed cells, whereas K63-linked polyubiquitinated TBK1 was hardly detected (**Fig. 5F**). Consistently, when we transfected EPC cells with TBK1-Myc, CDK2-HA, WT-Ub, K48-Ub, or K63-Ub, we found that CDK2 markedly increased K48-linked, but not K63-linked, ubiquitination of TBK1, while CDK2 knockdown remarkably attenuated K48-linked ubiquitination of TBK1 (**Fig. 5G** and **H**). Taken together, these findings indicate that the CDK2 triggers the degradation of TBK1 through K48-linked ubiquitination.

6. CDK2 recruits the ubiquitin E3 ligase Dtx4 to interact with and degrade TBK1

Since CDK2 is not an E3 ubiquitin ligase, it is speculated that CDK2-mediated degradation of TBK1 does not occur directly; instead, CDK2 acts as an adaptor to recruit an E3 ubiquitin ligase to TBK1. Several homologs of TBK1-associated E3 ubiquitin ligases in mammals were cloned into fish to detect protein interactions. It was found that Trim11 and Dtx4 are associated with TBK1, and these interactions were verified through inverse experiments (**Fig. 6A** and **B**). Moreover, protein interactions between CDK2 and Trim11 or Dtx4 were investigated. Dtx4 exhibited a significant interaction with CDK2 but not with Trim11, indicating that Dtx4 interacts with both TBK1 and CDK2 (**Fig. 6C**). Hence, Dtx4 was the main focus of the subsequent assays. Since Dtx4 is a potential ubiquitin ligase for TBK1, we verified whether CDK2 facilitated the interaction between Dtx4 and TBK1. Overexpression of CDK2 significantly increased the interaction between Dtx4 and TBK1, while CDK2 knockdown impeded this progress, suggesting that CDK2 promotes the interaction between Dtx4 and TBK1 (**Fig. 6D**). Next, the impact of the Dtx4 interaction was elucidated. Overexpression of DTX4 amplified CDK2-mediated inhibition of IFN promoter activity induced by TBK1 (**Fig. 6E**). Moreover, CDK2-mediated reduction of *ifn* and *vig1* mRNAs activated by TBK1 was also augmented by Dtx4 overexpression (**Fig. 6F**). In contrast to the normal state where CDK2 degrades TBK1, Dtx4 overexpression increased degradation (**Fig. 6G**). The degradation of endogenous TBK1 by CDK2 was more severe when Dtx4 was overexpressed (**Fig. 6H**). Two shRNAs were designed and generated, and after validation of exogenous and endogenous knockdown efficiencies, *shdtx4#1* was selected for the following assay (**Fig. 6I**). Dtx4 knockdown remarkably abrogated CDK2 suppression of TBK1-induced IFN promoter activity and *ifn* and *vig1* mRNAs, and CDK2 regulated both the exogenous and endogenous degradation of TBK1 (**Fig. 6J–M**). Overall, CDK2 recruits the E3 ubiquitin ligase Dtx4 to degrade TBK1.

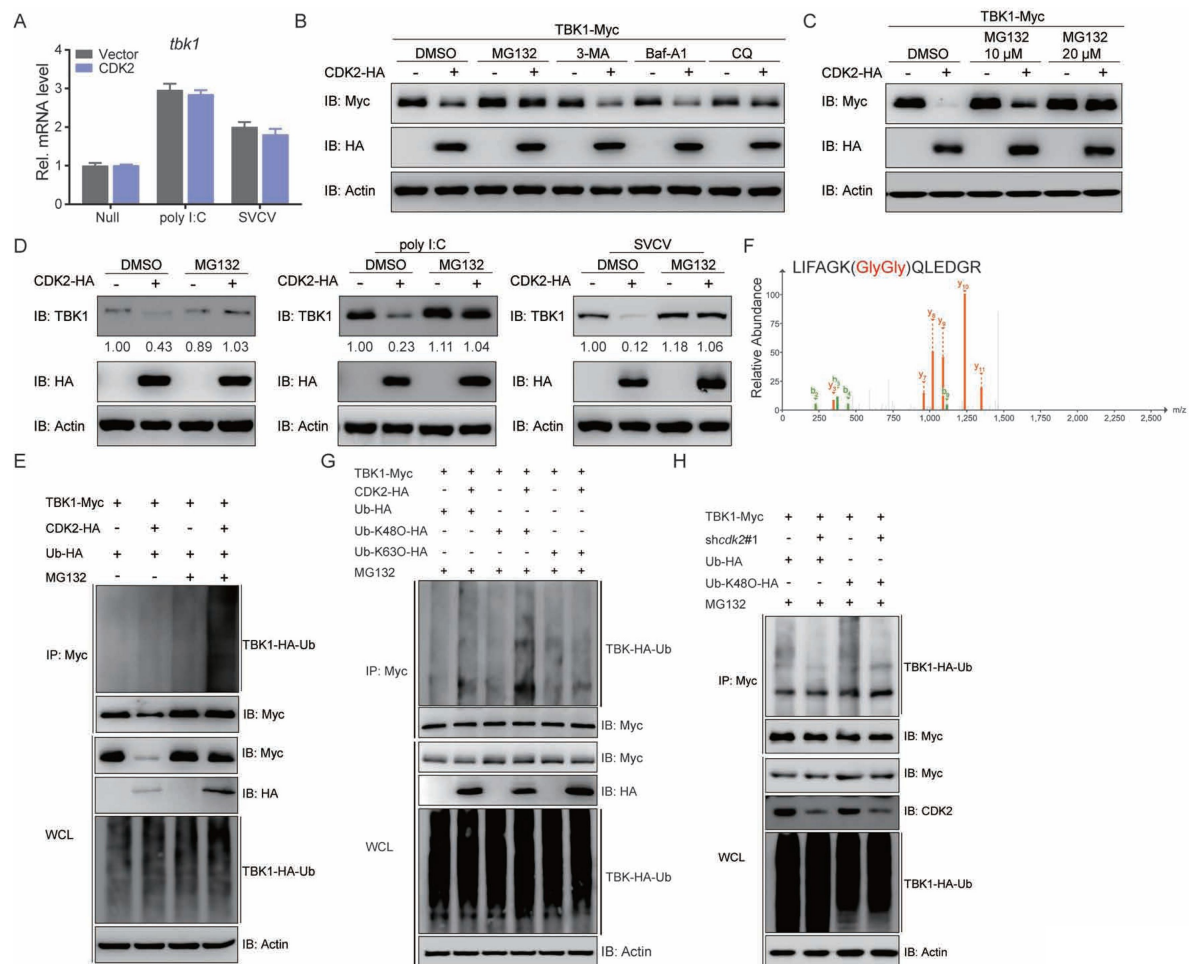


Figure 5.

CDK2 increases the K48-linked polyubiquitination of TBK1.

(A) qPCR analysis of *epc-tbk1* in EPC cells transfected with indicated plasmids for 24 h, and then untreated or infected with SVCV (MOI = 1) or transfected with poly I:C (2 μ g) for 24 h. (B and C) IB of proteins in EPC cells transfected with indicated plasmids for 18 h, followed by treatments of MG132 (10 μ M), 3-MA (2 mM), Baf-A1 (100 nM), and CQ (50 μ M) for 6 h, respectively. (D) IB of proteins in EPC cells transfected with CDK2-HA for 24 h, followed by untreated or infected with SVCV (MOI = 1) or transfected with poly I:C (2 μ g) for 24 h. Protein lysates were harvested after MG132 (20 μ M) treatments (6 h) for IB analysis. (E) TBK1 ubiquitination assays in EPC cells transfected with indicated plasmids for 18 h, followed by DMSO or MG132 treatments for 6 h. (F) Mass spectrometry analysis of a peptide derived from ubiquitinated TBK1-Myc. (G and H) TBK1 ubiquitination assays in EPC cells transfected with indicated plasmids for 18 h, followed by MG132 treatments for 6 h.

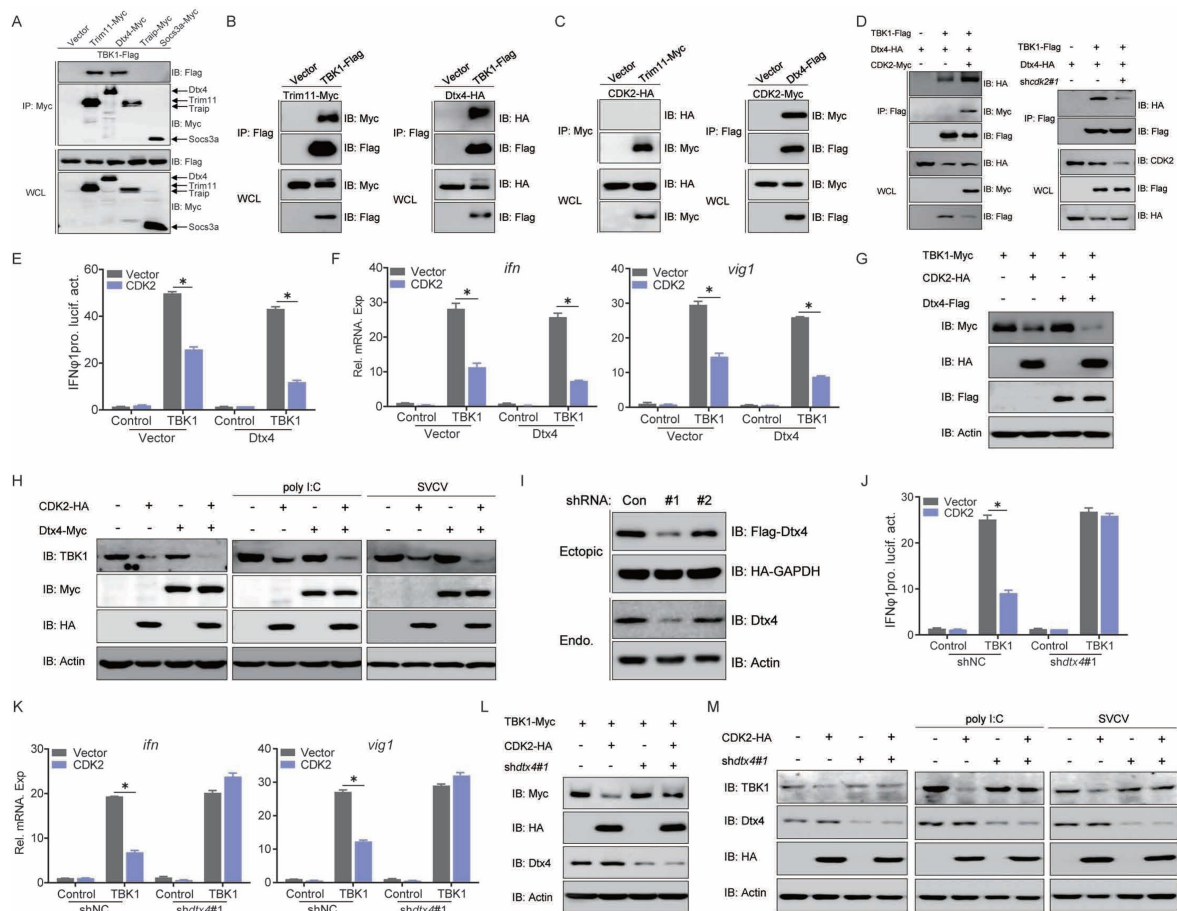


Figure 6.

CDK2 recruits Dtx4 to degrade TBK1.

(A-D) IB of WCLs and proteins immunoprecipitated with anti-Myc or Flag Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (E and J) Luciferase activity of IFN β 1pro in EPC cells transfected with indicated plasmids for 24 h. (F and K) qPCR analysis of *ifn* and *vig1* in EPC cells transfected with indicated plasmids for 24 h. (G and L) IB of proteins in EPC cells transfected with indicated plasmids for 24 h. (H and M) IB of proteins in EPC cells transfected with CDK2-HA and Dtx4-Myc or *shdtx4#1* for 24 h, followed by untreated or infected with SVCV (MOI = 1) or transfected with poly I:C (2 μ g) for 24 h.

7. The K567 site in TBK1 plays an essential role in CDK2-mediated degradation

By accelerating CDK2-mediated TBK1 degradation, the precise ubiquitin ligase function of Dtx4 was identified. Compared to Trim11, Dtx4 significantly enhanced the CDK2-potentiated ubiquitination of TBK1 (**Fig. 7A**). Further analysis revealed that Dtx4 overexpression augmented K48-linked TBK1 ubiquitination and that Dtx4 knockdown reduced this ubiquitination. Thus, demonstrating that Dtx4 is critical for CDK2-mediated TBK1 ubiquitination (**Fig. 7B** and **C**). To analyze the molecular mechanism of TBK1 modification by ubiquitination, two potential TBK1 lysine sites, namely K154 and K567, were identified for modification by ubiquitination using mass spectrometry analysis (**Fig. 7D**). Subsequently, two TBK1 mutants were generated with point mutations by mutating K154 and K567 to R to create K154R and K567R. In the ubiquitination assay, TBK1-K567R could not be ubiquitinated by CDK2, while K154R was almost identical to the wild-type TBK1 (**Fig. 7E**). This result was also characterized in the K48-linked ubiquitination assay (**Fig. 7F**). Moreover, Dtx4 significantly increased CDK2-mediated TBK1 wild-type and K48-linked ubiquitination but failed to enhance these processes in the K567R mutant group of TBK1 (**Fig. 7G** and **H**). This strongly suggests that K567 is a crucial site for TBK1 ubiquitination by Dtx4. Additionally, when TBK1-K567 was mutated, CDK2-mediated degradation of TBK1 failed, and the IFN induction and antiviral capacity restriction of TBK1 by CDK2 was also ineffective (**Fig. 7I–O**). These findings confirm the essential role of TBK1-K567 in recruiting Dtx4 through CDK2. The RING domain is crucial for E3 ubiquitin ligase activity. Three truncated mutants of Dtx4 were generated (**Fig. 7P**). The mutant lacking the RING domain (Dtx4- Δ RING) significantly impaired the interaction between Dtx4 and TBK1 or CDK2 (**Fig. 7Q**). In the ubiquitination assay, the wild-type Dtx4 enhanced TBK1 and K48-linked ubiquitination. On the other hand, Dtx4- Δ RING failed, indicating that the RING domain was necessary for TBK1 ubiquitination by Dtx4 (**Fig. 7R**). In summary, the data above demonstrates that the K567 site in TBK1 and the RING domain in the E3 ubiquitin ligase Dtx4 are crucial for CDK2 ubiquitination, leading to the degradation of TBK1.

Discussion

Although IFN responses exhibit powerful functions in defending against viral infection, excessive activation of IFN production may cause autoimmune disease. Therefore, the host needs to develop a set of regulatory mechanisms to balance immune responses. In this study, we illustrated a novel role for fish CDK2 in the negative regulation of IFN expression, which degraded TBK1 for K48-linked ubiquitination by recruiting E3 ubiquitin ligase Dtx4.

TBK1 is a central kinase in MAVS, STING, and TIR-domain-containing adapter-inducing IFN- β (TRIF) signaling complexes, which promote the phosphorylation of IRF3 and the production of IFN. Thus, the activation of TBK1 must be tightly regulated to avoid excessive autoimmune responses. TBK1 activity can be regulated in various ways, including phosphorylation, ubiquitination, SUMOylation, and preventing the formation of functional TBK1-containing complexes. For instance, receptor tyrosine kinase HER2 recruits AKT1 to directly phosphorylate TBK1, which disrupts the TBK1–STING association and K63-linked ubiquitination of TBK1, thus, suppressing antiviral responses¹⁸. Siglec1 associates with DAP12 and SHP2 to recruit the E3 ubiquitin ligase TRIM27, which induces K48-linked-ubiquitination-mediated TBK1 degradation, resulting in the inhibition of IFN production¹⁹. Similar to in mammals, TBK1 also dramatically activates the IFN signaling pathway in fish. Therefore, its activation must be precisely modulated. For example, cytokine receptor-like factor 3 (Crlf3) promotes the degradation of TBK1 via K48-linked ubiquitination, resulting in inhibition of IFN production²⁰. While TBK1 regulation has received attention, multiple molecules and underlying molecular mechanisms have not been fully characterized as potential TBK1 targets.

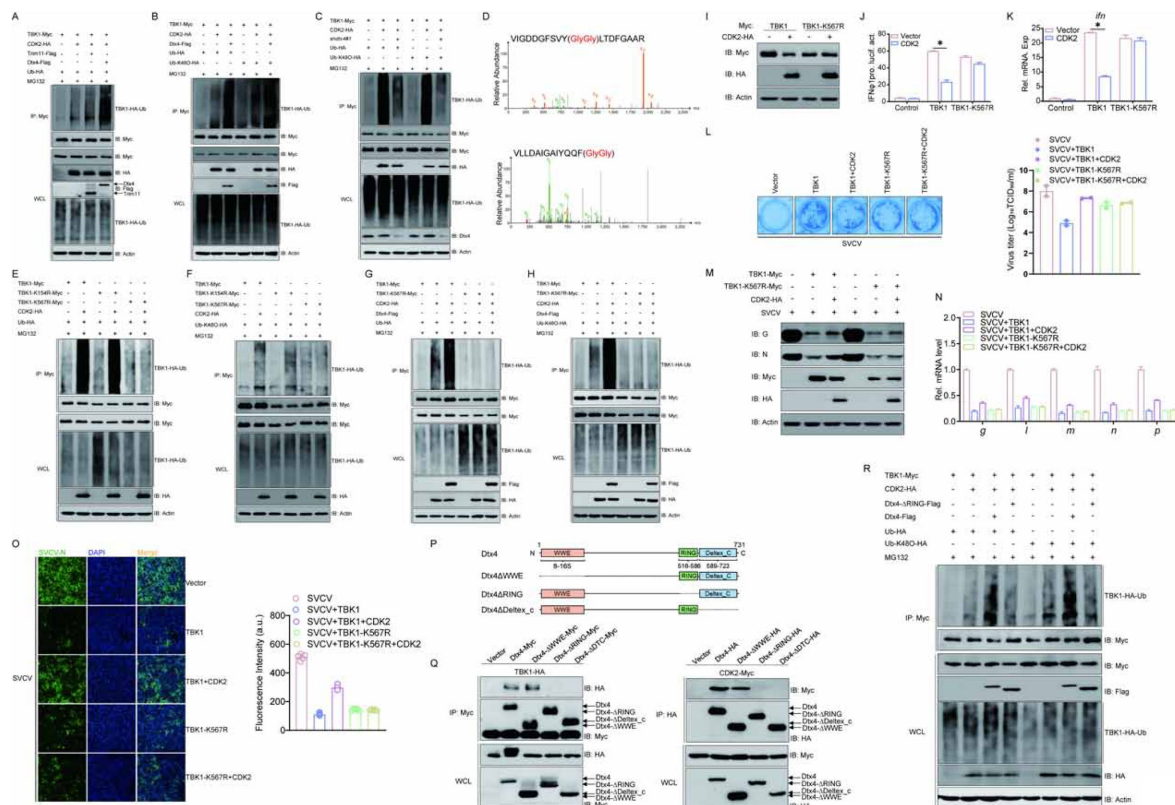


Figure 7.

K567 site is critical for the ubiquitination degradation of TBK1.

(A-C) TBK1 ubiquitination assays in EPC cells transfected with indicated plasmids for 18 h, followed by MG132 treatments for 6 h. (D) Mass spectrometry analysis to show that K154 and K567 of TBK1 is conjugated with ubiquitin. (E-H) TBK1 ubiquitination assays in EPC cells transfected with indicated plasmids for 18 h, followed by MG132 treatments for 6 h. (I) IB of proteins in EPC cells transfected with indicated plasmids for 24 h. (J) Luciferase activity of IFN β 1pro in EPC cells transfected with indicated plasmids for 24 h. (K) qPCR analysis of *ifn* in EPC cells transfected with indicated plasmids for 24 h. (L) Plaque assay of virus titers in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24-48 h. (M and N) qPCR and IB analysis of SVCV genes in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. (O) IF analysis of N protein in EPC cells transfected with indicated plasmids for 24 h, followed by SVCV challenge for 24 h. The fluorescence intensity (arbitrary unit, a.u.) was recorded by the LAS X software, and the data were expressed as mean $\pm$ SD, n = 5. (P) Schematic representation of full-length Dtx4 and its mutants. (Q) IB of WCLs and proteins immunoprecipitated with anti-Myc or HA Ab-conjugated agarose beads from EPC cells transfected with indicated plasmids for 24 h. (R) TBK1 ubiquitination assays in EPC cells transfected with indicated plasmids for 18 h, followed by MG132 treatments for 6 h.

Ubiquitination is one of the most versatile post-translational modifications (PTMs) of proteins and plays numerous vital roles in regulating antiviral responses. Ubiquitin comprises seven lysine residues (K6, K11, K27, K29, K33, K48, and K63); thus, seven different polyubiquitin chains can be produced²¹. K48-linked polyubiquitination is used to signal for proteasomal degradation of substrate proteins. In contrast, K63-linked polyubiquitination is a non-proteolytic mode of modification that plays several vital roles in stabilizing and activating target proteins²². Multiple studies have shown that different polyubiquitin chains can regulate the expression and activation of TBK1⁵. For instance, Nrdp1, MIB1/2, RNF128, and NLRC4 mediate the K63-linked polyubiquitination of TBK1 and facilitate its activation^{23, 24, 25, 26}. TRIP, SOCS3, and TRAF3IP3 have been proven to negatively regulate the IFN signaling pathway by targeting TBK1 for K48-linked polyubiquitination and degradation^{27, 28, 29}. Ubiquitination modification is a tightly regulated and reversible process that maintains cellular homeostasis. For example, the deubiquitinating enzymes CYLD, RNF114B, USP2b, and UBE2S remove the K63-linked polyubiquitination of TBK1 to block its activation^{30, 31, 32, 33}. Additionally, the USP1–UAF1 deubiquitinate complex has been found to cleave the K48-linked polyubiquitination of TBK1 to reverse its degradation process³⁴. These findings are specific to the ubiquitination of mammalian TBK1. Although fish TBK1 is highly conserved compared to mammalian TBK1, its ubiquitination modifications are worth exploring. Our study identified a previously unrecognized role for CDK2 in promoting the K48-linked polyubiquitination and proteasomal degradation of fish TBK1.

A series of ubiquitin-related enzymes are responsible for ubiquitination, including ubiquitin-activating enzymes (E1s), ubiquitin-conjugating enzymes (E2s), and ubiquitin ligases (E3s). Among them, the E3s are the critical components that determine the substrate specificity³⁵. E3s are generally divided into two large classes, including the homology to the E6-associated protein carboxyl terminus (HECT) domain-containing E3 ligases and the really interesting new gene (RING) domain-containing E3 ligases³⁶. Multiple studies have demonstrated that E3 ligases, including TRIP, Socs3, Dtx4, and TRIM27, specifically target TBK1 for K48-linked polyubiquitination and degradation^{19, 27, 28, 37}. However, CDK2 is not an E3 ubiquitin ligase. Thus, we reasoned that CDK2 might be a mediator in the recruitment of an E3 ubiquitin ligase for K48-linked polyubiquitination. To validate this hypothesis, we investigated and screened hTBK1-associated E3 ubiquitin ligases, demonstrating that Dtx4 interacted with CDK2 and enhanced the K48-linked polyubiquitination of TBK1. Meanwhile, our results showed that the RING domain in Dtx4 was necessary for modifying TBK1 through ubiquitination.

Recent studies have demonstrated that CDK activity is crucial for virus-induced innate immune responses³⁸. Reports indicate that CDKs are involved in the Toll-like receptor (TLR) signaling pathway, the nuclear factor- κ B (NF- κ B) signaling pathway, and the JAK-STAT signaling pathway. For instance, CDK8 and/or CDK19 enhanced the transcription of inflammatory genes, such as IL-8 and IL-10, in cells following TLR9 stimulation³⁹. CDKs and NF- κ B establish a remarkable paradigm where CDKs can act directly on substrate proteins rather than depending solely on transcriptional control⁴⁰. It has been reported that CDK1 serves as a positive regulator of the IFN-I signaling pathway, facilitating STAT1 phosphorylation, which subsequently boosts the expression of ISGs⁴¹. Furthermore, inhibiting CDK activity has been shown to obstruct STAT phosphorylation, proinflammatory gene activation, and ISG mRNA induction in response to SeV infection⁴². It is important to note that no evidence suggests the involvement of CDKs in RLR signaling pathways. This study has shown that fish CDK2 functions as a negative regulator of the key kinase TBK1, which is involved in the RLR signaling pathway. Variations in CDK2 activity during different phases of the cell cycle may lead to changes in the expression and function of TBK1. Our findings suggest that heightened CDK2 activity may suppress TBK1 expression, thereby hindering the cell's capacity to produce IFN. Conversely, during the late phase of the cell cycle or in an inhibited state, TBK1 expression may rise, enhancing IFN synthesis and release. In summary, CDK2 is involved in intracellular signaling by modulating TBK1 levels and IFN production, affecting the cellular immune response and cycle regulation—two processes that are notably

distinct at various stages of the cell cycle. A better understanding of the relationship between CDK2 and RLR signaling pathways will enhance our grasp of the regulatory mechanisms of CDKs in antiviral innate immunity.

CDK2 is a multifunctional kinase involved in many critical cellular processes, including cell cycle progression, differentiation, cancer, immunity, etc. ⁴³. To date, there has been limited research conducted on fish CDK2 in the regulation of cell cycle progression. The details are as follows: It has been reported that the kinase activity of goldfish CDK2 significantly increases during oocyte maturation ⁴⁴. Furthermore, UHRF1 phosphorylation by cyclin A2/CDK2 is crucial for zebrafish embryogenesis ⁴⁵. Additionally, a novel CDK2 homolog has been identified in Japanese lamprey, which plays a crucial role in apoptosis ⁴⁶. Red grouper nervous necrosis virus (RGNNV) infection activates the p53 pathway, leading to the upregulation of p21 and downregulation of cyclin E and CDK2, which forces infected cells to remain in the G1/S replicative phase ⁴⁷. In addition, we selected representative species from each of the six major vertebrate groups and compared their CDK2 protein sequences, discovering that they are over 90% similar to one another. This suggests that the function of CDK2 may be conserved to some extent across vertebrates. Furthermore, CDK2 inhibition has been shown to enhance anti-tumor immunity by increasing the IFN response to endogenous retroviruses ¹². Here, we reveal the role of CDK2 in modulating the RLR signaling pathway during innate immunity. Upon infection with SVCV, CDK2 expression was induced. CDK2 deficiency or knockdown enhanced the antiviral response both *in vitro* and *in vivo*, while CDK2 overexpression promoted viral replication. Thus, our study identifies CDK2 as a negative regulator of antiviral immune responses in addition to its well-studied function in cell cycle regulation. Meanwhile, the E3 ubiquitin ligase Dtx4 is also required to regulate antiviral immune responses. Further studies are needed to determine how CDK2 regulates Dtx4 activation and the detailed mechanisms involved.

In conclusion, our results identified a novel function of CDK2 in the negative regulation of TBK1-mediated IFN production. CDK2 interacted with TBK1 and recruited the E3 ubiquitin ligase Dtx4 to facilitate the K48-linked polyubiquitination at Lys567 residues in TBK1, eventually leading to the proteasomal degradation of TBK1. Our findings have revealed a previously unrecognized role for CDK2 in regulating immune homeostasis, providing molecular insight into the mechanisms through which CDK2-DTX4 targets TBK1 for ubiquitination and degradation.

Methods

Ethics statement

The experiments involved in this study were conducted in compliance with ethical regulations. The fish experiments were carried out under the guidance of the European Union Guidelines for the Handling of Laboratory Animals (2010/63/EU) and approved by the Ethics Committee for Animal Experiments of the Institute of Aquatic Biology of the Chinese Academy of Sciences (No. 2023-068).

Fish, cells, and viruses

Mature zebrafish individuals 2.5 months after hatching (0.4 ± 0.1 g) were selected in this study. AB line wild-type zebrafish (*Danio rerio*) and *cdk2* mutant zebrafish (CZ1442: *cdk2*^{ihb488/+}) were obtained from the China National Zebrafish Resource Center (CZRC) and bred using standardized procedures. In accordance with ethical requirements and national animal welfare guidelines, all experimental fish were required to undergo a two-week acclimatization period in the laboratory and have their health assessed prior to the study. Only fish that appeared healthy and were mobile were used for scientific research. ZF4 cells (American Type Culture Collection, ATCC) were cultured in Ham's F-12 medium (Invitrogen) supplemented with 10% fetal bovine serum (FBS) at 28 °C and 5% CO₂. EPC cells and ctenopharyngodon idellus kidney (CIK) cells were obtained from

the Chinese Culture Collection Centre for Type Cultures (CCTCC), Gibel carp brain (GiCB) cells were provided by Ling-Bing Zeng (Yangtze River Fisheries Research Institute, Chinese Academy of Fishery Sciences), these cells were maintained at 28°C in 5% CO₂ in medium 199 (Invitrogen) supplemented with 10% FBS. THP1 cells originally obtained from ATCC were maintained in RPMI 1640 medium supplemented with 10% FBS. SVCV was propagated in EPC cells until a cytopathogenic effect (CPE) was observed, and then cell culture fluid containing SVCV was harvested and centrifuged at $4 \times 10^3 g$ for 20 min to remove the cell debris, and the supernatant was stored at -80°C until used. GCRV (strain 873, group I) and vesicular stomatitis virus (VSV) was provided by Prof. Wuhan Xiao (Institute of Hydrobiology, Chinese Academy of Sciences). GCRV was propagated in CIK cells and harvested in a similar way to SVCV. Cyprinid herpesvirus 2 (CyHV2, obtained from Yancheng city, Jiangsu province, China) was provided by Prof. Liqun Lu (Shanghai Ocean University). CyHV-2 was propagated in GiCB cells and harvested in a similar way to SVCV.

Plasmid construction and reagents

The sequence of zebrafish CDK2 (GenBank accession number: NM_213406.1) was obtained from the National Centre for Biotechnology Information (NCBI) website. CDK2 was amplified by polymerase chain reaction (PCR) using cDNA from adult zebrafish tissues as a template and cloned into the expression vector pCMV-HA or pCMV-Myc (Clontech) vectors. Zebrafish MAVS (NM_001080584.2), TBK1 (NM_001044748.2) and the truncated mutants of TBK1, Dtx4 (XM_002660524.5) and the truncated mutants of Dtx4, Trim11 (XM_021470074.1), Traip (NM_205607.1), Socs3a (NM_199950.1), and GAPDH (NM_001115114.1) were cloned into pCMV-Myc and pCMV-Tag2C vectors. The short hairpin RNA of *Pimephales promelas* CDK2 (XM_039663387.1) and Dtx4 (XM_039689084.1) were designed by BLOCK-iT RNAi Designer and cloned into the pLKO.1-TRC Cloning vector. For subcellular localization experiments, CDK2 was constructed onto pEGFP-N3 (Clontech), while MAVS and TBK1 were constructed onto pCS2-mCherry (Clontech). The plasmids containing zebrafish IFN ϕ 1pro-Luc and ISRE-Luc in the pGL3-Basic luciferase reporter vector (Promega) were constructed as described previously. The *Renilla* luciferase internal control vector (pRL-TK) was purchased from Promega. The ubiquitin mutant expression plasmids Lys-48 and Lys-63 (all lysine residues were mutated except Lys-48 or Lys-63) were ligated into the pCMV-HA vectors named Ub-K48O-HA and Ub-K63O-HA. All constructs were confirmed by DNA sequencing. Polyinosinic-polycytidylic acid (poly I:C) was purchased from Sigma-Aldrich used at a final concentration of 1 $\mu g/\mu l$. MG132 (Cat. No. M7449), 3-Methyladenine (3-MA) (Cat. No. M9281), Chloroquine (CQ) (Cat. No. C6628) were obtained from Sigma-Aldrich. Bafilomycin A1 (Baf-A1) (Cat. No. S1413) was obtained from Selleck.

Transcriptomic analysis

Total RNA was extracted using the TRIzol method and assessed for RNA purity and quantification using a NanoDrop 2000 spectrophotometer (Thermo Scientific, Waltham, U.S.A.), and RNA integrity was assessed using an Agilent 2100 Bioanalyzer (Agilent Technologies, Santa Clara, U.S.A.). Santa Clara, U.S.A.). Transcriptome sequencing and data analysis were performed by OE Biotech (Shanghai, China). The raw sequencing data was submitted to the NGDC (National Genomics Data Center) (GSA accession number: CRA008409).

Transient transfection and virus infection

EPC cells were transfected in 6-well and 24-well plates using transfection reagents from FishTrans (MeiSenTe Biotechnology) according to the manufacturer's protocol. Antiviral assays were performed in 24-well plates by transfecting EPC cells with the plasmids shown in the figure. At 24 h post-transfection, cells were infected with SVCV at a multiplicity of infection (MOI = 0.001), GCRV (MOI = 0.001), CyHV-2 (MOI = 0.1). After 48 h or 72 h, supernatant aliquots were harvested for detection of virus titers, the cell monolayers were fixed by 4% paraformaldehyde (PFA) and stained with 1% crystal violet for visualizing CPE. For virus titration, 200 μl of culture medium were collected at 48 h post-infection and used for detection of virus titers according to the method

of Reed and Muench. The supernatants were subjected to 3-fold or 10-fold serial dilutions and then added (100 μ l) onto a monolayer of EPC cells cultured in a 96-well plate. After 48 or 72 h, the medium was removed and the cells were washed with PBS, fixed by 4% PFA and stained with 1% crystal violet. The virus titer was expressed as 50% tissue culture infective dose (TCID₅₀/ml). For viral infection, fish were anesthetized with methanesulfonate (MS-222) and intraperitoneally (i.p.) injected with 5 μ l of M199 containing SVCV (5×10^8 TCID₅₀/ml). The i.p. injection of PBS was used as mock infection. Then the fish were migrated into the aquarium containing new aquatic water.

Luciferase activity assay

EPC cells were cultured overnight in 24-well plates and then co-transfected with the expression plasmid and luciferase reporter plasmid. The cells were infected with SVCV or transfected with poly I:C for 24 h prior to harvest. To ensure that the same total amount of DNA was transfected in each well, pCMV-HA empty vector was used. At 24 h post-stimulation, cells were washed with phosphate-buffered saline (PBS) and lysed for measuring luciferase activity by the Dual-Luciferase Reporter Assay System (Promega) according to the manufacturer's instructions. Firefly luciferase activity was normalized based on the Renilla luciferase activity.

RNA extraction, reverse transcription, and quantitative PCR (qPCR)

The RNA was extracted using TRIzol reagent (Invitrogen), and first-strand cDNA was synthesized with a PrimeScript RT kit with gDNA Eraser (Takara). qPCR was performed on the CFX96 Real-Time System (Bio-Rad) using SYBR green PCR Master Mix (Yeast). PCR conditions were as follows: 95°C for 5 min and then 40 cycles of 95°C for 20 s, 60°C for 20 s, and 72°C for 20 s. All primers used for qPCRs are shown in Table S1, and β -actin gene was used as an internal control. The relative fold changes were calculated by comparison to the corresponding controls using the $2^{-\Delta\Delta C_t}$ method (where C_t was the threshold cycle).

Co-immunoprecipitation (Co-IP) assay

EPC cells were cultured in 10 cm² dishes overnight and transfected with 10 μ g of plasmid, as shown. At 24 h post-transfection, the medium was discarded, and the cells were washed with PBS. Then the cells were lysed in 1 ml of radioimmunoprecipitation (RIPA) lysis buffer [1% NP-40, 50 mM Tris-HCl, pH 7.5, 150 mM NaCl, 1 mM EDTA, 1 mM NaF, 1 mM sodium orthovanadate (Na₃VO₄), 1 mM phenyl-methylsulfonyl fluoride (PMSF), 0.25% sodium deoxycholate] containing protease inhibitor cocktail (Sigma-Aldrich) at 4°C for 1 h on a rocker platform. The cellular debris was removed by centrifugation at 12,000 $\times g$ for 15 min at 4°C. The supernatant was transferred to a fresh tube and incubated with 20 μ l anti-Flag/HA/Myc affinity gel (Sigma-Aldrich) overnight at 4°C with constant rotating incubation. These samples were further analyzed by immunoblotting (IB). Immunoprecipitated proteins were collected by centrifugation at 5000 $\times g$ for 1 min at 4°C, washed three times with lysis buffer and resuspended in 50 μ l 2 $\times$ SDS sample buffer. The immunoprecipitates and whole cell lysates (WCLs) were analyzed by IB with the indicated antibodies (Abs).

In vivo ubiquitination assay

Transfected EPC cells were washed twice with 10 mL ice-cold PBS and then digested with 1 mL 0.25% trypsin-EDTA (1 $\times$) (Invitrogen) for 2–3 min until the cells were dislodged. 100 μ l FBS was added to neutralize the trypsin and the cells were resuspended into 1.5 mL centrifuge tube, centrifuged at 2000 $\times g$ for 5 min. The supernatant was discarded and the cell precipitations were resuspended using 1 mL PBS and centrifuged at 2000 $\times g$ for 5 min. The collected cell precipitations were lysed using 100 μ l PBS containing 1% SDS and denatured by heating for 10 min. The supernatants were diluted with lysis buffer until the concentration of SDS was decreased to 0.1%. The diluted supernatants were incubated with 20 μ l anti-Myc affinity gel overnight at 4°C with

constant agitation. These samples were further analyzed by IB. Immunoprecipitated proteins were collected by centrifugation at 5000 $\times g$ for 1 min at 4°C, washed for three times with lysis buffer and resuspended in 100 μ L 1 $\times$ SDS sample buffer.

Immunoblot analysis

Immunoprecipitates or WCLs were analyzed as described previously. Antibodies were diluted as follows: anti- β -actin (ABclonal, AC026) at 1:3000, anti-Flag (Sigma-Aldrich, F1804) at 1:3000, anti-HA (Covance, MMS-101R) at 1:3000, anti-Myc (Santa Cruz Biotechnology, sc-40) at 1:3000, anti-Dtx4 (Thermo Scientific, PA5-46146) at 1:1000, anti-CDK2 (GeneTex, GTX101226) at 1:1000, and HRP-conjugated anti-mouse/rabbit IgG (Thermo Scientific, 31430/31460) at 1:5000, anti-N/P/G/TBK1/CDK2 (prepared and purified in our lab) at 1:2000.

Immunofluorescence (IF)

EPC cells were plated onto glass coverslips in 6-well plates and infected with SVCV (MOI = 1) 24 h. Then the cells were washed with PBS and fixed in 4% PFA at room temperature for 1 h and permeabilized with 0.2% Triton X-100 in ice-cold PBS for 15 min. The samples were blocked for 1 h at room temperature in PBS containing 2% bovine serum albumin (BSA, Sigma-Aldrich). After additional PBS washing, the samples were incubated with anti-N Ab in PBS containing 2% BSA for 2–4 h at room temperature. After three times washed by PBS, the samples were incubated with secondary Ab (Alexa Fluor 488 AffiniPure Donkey anti-Rabbit IgG (H+L) (34206ES60, 1:10,000) in PBS containing 2% BSA for 1 h at room temperature. After additional PBS washing, the cells were finally stained with 1 μ g/ml 4', 6-diamidino-2-phenylindole (DAPI; Beyotime Institute of Biotechnology, Shanghai, China) for 10 min in the dark at room temperature. Finally, the coverslips were washed and observed with a confocal microscope under a 10 $\times$ immersion objective (SP8; Leica).

Fluorescent microscopy

EPC cells were plated onto coverslips in 6-well plates and transfected with the plasmids indicated for 24 h. Then the cells were washed twice with PBS and fixed with 4% PFA for 1 h. After being washed three times with PBS, the cells were stained with 1 μ g/ml DAPI for 15 min in the dark at room temperature. Finally, the coverslips were washed and observed with a confocal microscope under a 63 $\times$ oil immersion objective (SP8; Leica).

Histopathology

Liver, spleen, and kidney tissues from three individuals of control or virus-infected fish at 2 days post infection (dpi) were dissected, and fixed in 10% phosphate-buffered formalin overnight. Then the samples were dehydrated in ascending grades of alcohol and embedded into paraffin. Sections at 5 μ m thickness were taken and stained with hematoxylin and eosin (H&E). Histological changes were examined by optical microscopy at $\times 40$ magnification and were analyzed by the Aperio ImageScope software.

Statistics analysis

For fish survival analysis, Kaplan-Meier survival curves were generated and analyzed by Log-rank test. For the bar graph, one representative experiment of at least three independent experiments is shown, and each was done in triplicate. For the dot plot graph, each dot point represents one independent biological replicate. Unpaired Student's *t* test was used for statistical analysis. Data are expressed as mean $\pm$ standard error of the mean (SEM). A *p* value < 0.05 was considered statistically significant.

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Additional information

Author Contributions

L.F.L. and S.L. conceptualized the study and wrote the original draft. L.F.L. did data analysis. C.Z., Z.C.L., and B.J.C. were responsible for sample collection. D.D.C., S.L., and L.F.L. are responsible for funding resources. All authors reviewed and edited the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

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Author information

Long-Feng Lu

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Can Zhang

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Zhuo-Cong Li

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Bao-Jie Cui

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, College of Fisheries and Life Science, Dalian Ocean University, Dalian, China

Yang-Yang Wang

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Ke-Jia Han

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, College of Fisheries and Life Science, Dalian Ocean University, Dalian, China

Xiao Xu

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, College of Fisheries and Life Science, Dalian Ocean University, Dalian, China

Chu-Jing Zhou

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, College of Fisheries and Life Science, Dalian Ocean University, Dalian, China

Xiao-Yu Zhou

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Yue Wu

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Na Xu

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Xiao-Li Yang

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Dan-Dan Chen

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China

Xi-Yin Li

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China, State Key Laboratory of Freshwater Ecology and Biotechnology, Institute of Hydrobiology, Wuhan, China, Key Laboratory of Aquaculture Disease Control, Ministry of Agriculture, Wuhan, China

Li Zhou

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, University of Chinese Academy of Sciences, Beijing, China, State Key Laboratory of Freshwater Ecology and Biotechnology, Institute of Hydrobiology, Wuhan, China, Key Laboratory of Aquaculture Disease Control, Ministry of Agriculture, Wuhan, China

Shun Li

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China, Laboratory for Marine Biology and Biotechnology, Qingdao Marine Science and Technology Center, Qingdao, China, University of Chinese Academy of Sciences, Beijing, China, State Key Laboratory of Freshwater Ecology and Biotechnology, Institute of Hydrobiology, Wuhan, China, Key Laboratory of Aquaculture Disease Control, Ministry of Agriculture, Wuhan, China
ORCID iD: [0000-0002-3629-9900](https://orcid.org/0000-0002-3629-9900)

For correspondence: bob@ihb.ac.cn

Editors

Reviewing Editor

Larissa Cunha

University of Sao Paulo, Ribeirão Preto, Brazil

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Tadatsugu Taniguchi

University of Tokyo, Tokyo, Japan

Reviewer #1 (Public review):

Summary:

The authors set out to evaluate the regulation of interferon (IFN) gene expression in fish, using mainly zebrafish as a model system. Similar to more widely characterized mammalian systems, fish IFN is induced during viral infection through the action of the transcription factor IRF3 which is activated by phosphorylation by the kinase TBK1. It has been previously shown in many systems that TBK1 is subjected to both positive and negative regulation to control IFN production. In this work, the authors find that the cell cycle kinase CDK2 functions as a TBK1 inhibitor by decreasing its abundance through recruitment of the ubiquitylation ligase, Dtx4, which has been similarly implicated in the regulation of mammalian TBK1. Experimental data are presented showing that CDK2 interacts with both TBK1 and Dtx4, leading to TBK1 K48 ubiquitylation on K567 and its subsequent degradation by the proteasome.

Strengths:

The strengths of this manuscript are its novel demonstration of the involvement of CDK2 in a process in fish that is controlled by different factors in other vertebrates and its clear and supportive experimental data.

Weaknesses:

The weaknesses of the study include the following. 1) It remains unclear how CDK is regulated during viral infection and how it specifically recruits E3 ligase to TBK1. 2) The implications and mechanisms for a relationship between the cell cycle and IFN production will be a fascinating topic for future studies.

<https://doi.org/10.7554/eLife.98357.2.sa2>

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<https://doi.org/10.7554/eLife.98357.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):**Weaknesses:**

The weaknesses of the study include the following.

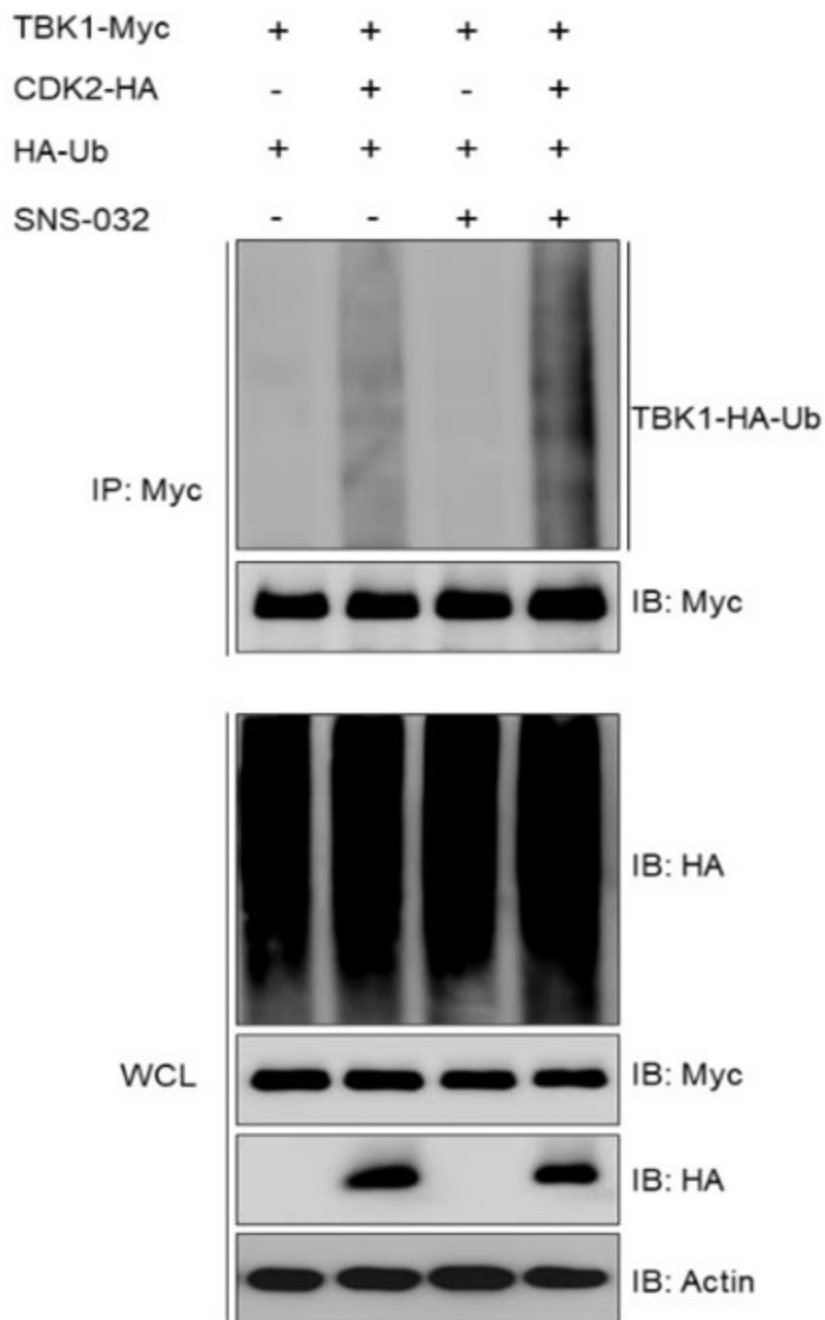
(1) It remains unclear whether the function described for CDK2 is regulatory, that is, it affects TBK1 levels during physiological responses such as viral infection or cell cycle progression, or if it is homeostatic, governing the basal abundance of TBK1 but not responding to signaling.

The regulation of TBK1 by CDK2 described in this article occurs during viral infection. Simultaneously, we also investigated the effects of CDK2 overexpression and knockdown on TBK1 levels under non-infected state and observed a slight reduction, as shown in Figure 4K and 4L. Thus, we speculate that the regulation of TBK1 by CDK2 serves, on one hand, to maintain cellular homeostasis and, on the other hand, to respond to signaling triggered by viral infection.

(2) The authors have not explored whether the catalytic activity of CDK2 is required for TBK1 ubiquitinylation and, if so, what its target specificity is.

We found that the ubiquitination modification of TBK1 was not affected by treatment with a CDK2 kinase activity inhibitor (SNS-032), as demonstrated in the results below (Author response image 1).

Author response image 1.



(3) Given the multitude of CDK isoforms in fish, it remains unexplored whether the identified fish CDK2 homolog is a requisite cell cycle regulator or if its action in the cell cycle is redundant with other CDKs.

A comparison of the protein sequences of fish CDK2 and human CDK2 revealed a 90% similarity (Author response image 2). It has also been reported that the kinase activity of goldfish CDK2 significantly increases during oocyte maturation (ref. 1). Furthermore, UHRF1 phosphorylation by cyclin A2/CDK2 is crucial for zebrafish embryogenesis (ref. 2). Additionally, Red grouper nervous necrosis virus (RGNNV) infection activated the p53 pathway, leading to the upregulation of p21 and downregulation of cyclin E and CDK2, which

forces infected cells to remain in the G1/S replicative phase (ref. 3). All these evidences suggest that fish CDK2 plays a vital role in cell cycle regulation, and there have been no reports of other CDKs demonstrating CDK2-like functions.

References:

- (1) Hirai T, et al. (1992) Isolation and Characterization of Goldfish Cdk2, a Cognate Variant of the Cell-Cycle Regulator Cdc2. *Developmental biology* 152(1):113-120.
- (2) Chu J, et al. (2012) UHRF1 phosphorylation by cyclin A2/cyclin-dependent kinase 2 is required for zebrafish embryogenesis. *Molecular biology of the cell* 23(1):59-70.
- (3) Mai WJ, Liu HX, Chen HQ, Zhou YJ, & Chen Y (2018) RGNNV-induced cell cycle arrest at G1/S phase enhanced viral replication via p53-dependent pathway in GS cells. *Virus Res* 256:142-152.

Author response image 2.

Homo_sapiens.seq	MEFQKVEKIGEGTYGVVYKANKITGEVVALKKIRLDTE	40
Danio_rerio.seq	MEFQKVEKIGEGTYGVVYKANKITGEVVALKKIRLDTE	40
Pimephales_promelas.seq	MEFQKVEKIGEGTYGVVYKANKITGEVVALKKIRLDTE	40
Ctenopharyngodon_idella.seq	MEFQKVEKIGEGTYGVVYKANKITGEVVALKKIRLDTE	40
Carassius_auratus.seq	MEFQKVEKIGEGTYGVVYKANKITGEVVALKKIRLDTE	40
Consensus	MEFQKVEKIGEGTYGVVYKANKITGEVVALKKIRLDTE	
Homo_sapiens.seq	TEGVPTAIREISLLKELNHPNIVKLDVIHTENKLYLVF	80
Danio_rerio.seq	TEGVPTAIREISLLKELNHPNIVKLDVIHTENKLYLVF	80
Pimephales_promelas.seq	TEGVPTAIREISLLKELNHPNIVKLDVIHTENKLYLVF	80
Ctenopharyngodon_idella.seq	TEGVPTAIREISLLKELNHPNIVKLDVIHTENKLYLVF	80
Carassius_auratus.seq	TEGVPTAIREISLLKELNHPNIVKLDVIHTENKLYLVF	80
Consensus	TEGVPTAIREISLLKELNHPNIVKLDVIHTENKLYLVF	
Homo_sapiens.seq	EFLHCDLKFMEFASALIGISLPLKSYLFQLLQGLAFCHS	120
Danio_rerio.seq	EFLHCDLKFMEFASALIGISLPLKSYLFQLLQGLAFCHS	120
Pimephales_promelas.seq	EFLHCDLKFMEFASALIGISLPLKSYLFQLLQGLAFCHS	120
Ctenopharyngodon_idella.seq	EFLHCDLKFMEFASALIGISLPLKSYLFQLLQGLAFCHS	120
Carassius_auratus.seq	EFLHCDLKFMEFASALIGISLPLKSYLFQLLQGLAFCHS	120
Consensus	EFLHCDLKFMEFASALIGISLPLKSYLFQLLQGLAFCHS	
Homo_sapiens.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVPTTYT	160
Danio_rerio.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVPTTYT	160
Pimephales_promelas.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVPTTYT	160
Ctenopharyngodon_idella.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVPTTYT	160
Carassius_auratus.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVPTTYT	160
Consensus	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVPTTYT	
Homo_sapiens.seq	HEVVTILWYRAPEILLGCKYYSTAVDIWSLGCIFAEMVTR	200
Danio_rerio.seq	HEVVTILWYRAPEILLGCKYYSTAVDIWSLGCIFAEMVTR	200
Pimephales_promelas.seq	HEVVTILWYRAPEILLGCKYYSTAVDIWSLGCIFAEMVTR	200
Ctenopharyngodon_idella.seq	HEVVTILWYRAPEILLGCKYYSTAVDIWSLGCIFAEMVTR	200
Carassius_auratus.seq	HEVVTILWYRAPEILLGCKYYSTAVDIWSLGCIFAEMVTR	200
Consensus	HEVVTILWYRAPEILLGCKYYSTAVDIWSLGCIFAEMVTR	
Homo_sapiens.seq	ALFPGDSEIDQLFRIFRTLGTPEDESWPGVTSMPDYKPSF	240
Danio_rerio.seq	ALFPGDSEIDQLFRIFRTLGTPEDESWPGVTSMPDYKPSF	240
Pimephales_promelas.seq	ALFPGDSEIDQLFRIFRTLGTPEDESWPGVTSMPDYKPSF	240
Ctenopharyngodon_idella.seq	ALFPGDSEIDQLFRIFRTLGTPEDESWPGVTSMPDYKPSF	240
Carassius_auratus.seq	ALFPGDSEIDQLFRIFRTLGTPEDESWPGVTSMPDYKPSF	240
Consensus	ALFPGDSEIDQLFRIFRTLGTPEDESWPGVTSMPDYKPSF	
Homo_sapiens.seq	PKWARCEISKVVPFLDGDGRLLSCMLHYDPNKRISAKAA	280
Danio_rerio.seq	PKWARCEISKVVPFLDGDGRLLSCMLHYDPNKRISAKAA	280
Pimephales_promelas.seq	PKWARCEISKVVPFLDGDGRLLSCMLHYDPNKRISAKAA	280
Ctenopharyngodon_idella.seq	PKWARCEISKVVPFLDGDGRLLSCMLHYDPNKRISAKAA	280
Carassius_auratus.seq	PKWARCEISKVVPFLDGDGRLLSCMLHYDPNKRISAKAA	280
Consensus	PKWARCEISKVVPFLDGDGRLLSCMLHYDPNKRISAKAA	
Homo_sapiens.seq	LAHFFFDVTPVPELR	297
Danio_rerio.seq	LAHFFFDVTPVPELR	297
Pimephales_promelas.seq	LAHFFFDVTPVPELR	297
Ctenopharyngodon_idella.seq	LAHFFFDVTPVPELR	297
Carassius_auratus.seq	LAHFFFDVTPVPELR	297
Consensus	LAHFFFDVTPVPELR	

Reviewer #2 (Public Review):

Weaknesses:

- (1) While the study focuses on fish, the broader implications for other lower vertebrates and higher vertebrates are not extensively discussed.

Thanks to your comment, we have added a paragraph to the Discussion section of the manuscript regarding the implications of the negative regulation of IFN expression by fish CDK2 for other vertebrates (lines 398-403). The details are as follows: first, we selected representative species from each of the six major vertebrate groups and compared their CDK2 protein sequences, finding that they are over 90% similar to one another (Author response image 3). This suggests that the function of CDK2 may be conserved to some extent across vertebrates. Additionally, CDK2 inhibition has been shown to enhance anti-tumor immunity by increasing the IFN response to endogenous retroviruses (ref. 1). Our studies provide evidence that fish CDK2 inhibits the IFN response by promoting the ubiquitination and degradation of TBK1, strongly supporting the role of CDK2 in the regulation of the immune response.

Reference:

(1) Chen Y, et al. (2022) CDK2 Inhibition Enhances Antitumor Immunity by Increasing IFN Response to Endogenous Retroviruses. *Cancer Immunol Res* 10(4):525-539.

Author response image 3.

Homo_sapiens.seq	MENFQKVEKIGEGTYGVVYKARNITGEVVALKKIRLDTE	40
Gallus_gallus-CDK2.seq	MENFQKVEKIGEGTYGVVYKARNITGEVVALKKIRLDTE	40
Sceloporus_undulatus_-CDK2.seq	MENFQKVEKIGEGTYGVVYKARNITGEVVALKKIRLDTE	40
Xenopus_laevis-CDK2.seq	MENFQKVEKIGEGTYGVVYKARNITGEVVALKKIRLDTE	40
Pimephales_promelas.seq	MENFQKVEKIGEGTYGVVYKARNITGEVVALKKIRLDTE	40
Ctenopharyngodon_idella.seq	MENFQKVEKIGEGTYGVVYKARNITGEVVALKKIRLDTE	40
Carassius_auratus.seq	MENFQKVEKIGEGTYGVVYKARNITGEVVALKKIRLDTE	40
Danio_rerio.seq	MENFQKVEKIGEGTYGVVYKARNITGEVVALKKIRLDTE	40
Consensus	me f q k v e k i g e g t y g v v y k a n i t g e v v a l k k i r l d t e	
Homo_sapiens.seq	TEGVVPSTAIAREISLLKELNHPNIVKLRDVIHTENKLYLVF	80
Gallus_gallus-CDK2.seq	TEGVVPSTAIAREISLLKELNHPNIVKLRDVIHTENKLYLVF	80
Sceloporus_undulatus_-CDK2.seq	TEGVVPSTAIAREISLLKELNHPNIVKLRDVIHTENKLYLVF	80
Xenopus_laevis-CDK2.seq	TEGVVPSTAIAREISLLKELNHPNIVKLRDVIHTENKLYLVF	80
Pimephales_promelas.seq	TEGVVPSTAIAREISLLKELNHPNIVKLRDVIHTENKLYLVF	80
Ctenopharyngodon_idella.seq	TEGVVPSTAIAREISLLKELNHPNIVKLRDVIHTENKLYLVF	80
Carassius_auratus.seq	TEGVVPSTAIAREISLLKELNHPNIVKLRDVIHTENKLYLVF	80
Danio_rerio.seq	TEGVVPSTAIAREISLLKELNHPNIVKLRDVIHTENKLYLVF	80
Consensus	te g v p s t a i a r e i s l l k e l n h p n i v k l r d v i h t e n k l y l v f	
Homo_sapiens.seq	EFLQDILKRFMEASASLIGISLILVKSYLEFQLLQGLAFCH	119
Gallus_gallus-CDK2.seq	EFLQDILKRFMEASASLIGISLILVKSYLEFQLLQGLAFCH	119
Sceloporus_undulatus_-CDK2.seq	EFLQDILKRFMEASASLIGISLILVKSYLEFQLLQGLAFCH	120
Xenopus_laevis-CDK2.seq	EFLQDILKRFMEASASLIGISLILVKSYLEFQLLQGLAFCH	119
Pimephales_promelas.seq	EFLQDILKRFMEASASLIGISLILVKSYLEFQLLQGLAFCH	119
Ctenopharyngodon_idella.seq	EFLQDILKRFMEASASLIGISLILVKSYLEFQLLQGLAFCH	119
Carassius_auratus.seq	EFLQDILKRFMEASASLIGISLILVKSYLEFQLLQGLAFCH	119
Danio_rerio.seq	EFLQDILKRFMEASASLIGISLILVKSYLEFQLLQGLAFCH	119
Consensus	e f l q d i l k r f m e a s a s l i g i s l i l v k s y l e f q l l q g l a f c h	
Homo_sapiens.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVVRT	159
Gallus_gallus-CDK2.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVVRT	159
Sceloporus_undulatus_-CDK2.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVVRT	160
Xenopus_laevis-CDK2.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVVRT	159
Pimephales_promelas.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVVRT	159
Ctenopharyngodon_idella.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVVRT	159
Carassius_auratus.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVVRT	159
Danio_rerio.seq	HRVLHRLDKPQNLLINAGCEIKLADFGLARAFGVVVRT	159
Consensus	h r v l h r d l k p q n l l i n a g c e i k l a d f g l a r a f g v v v r t	
Homo_sapiens.seq	THEVVTILWYRAPEILLGCKYSTAVDIWSLGCIFAEMVTR	199
Gallus_gallus-CDK2.seq	THEVVTILWYRAPEILLGCKYSTAVDIWSLGCIFAEMVTR	199
Sceloporus_undulatus_-CDK2.seq	THEVVTILWYRAPEILLGCKYSTAVDIWSLGCIFAEMVTR	200
Xenopus_laevis-CDK2.seq	THEVVTILWYRAPEILLGCKYSTAVDIWSLGCIFAEMVTR	199
Pimephales_promelas.seq	THEVVTILWYRAPEILLGCKYSTAVDIWSLGCIFAEMVTR	199
Ctenopharyngodon_idella.seq	THEVVTILWYRAPEILLGCKYSTAVDIWSLGCIFAEMVTR	199
Carassius_auratus.seq	THEVVTILWYRAPEILLGCKYSTAVDIWSLGCIFAEMVTR	199
Danio_rerio.seq	THEVVTILWYRAPEILLGCKYSTAVDIWSLGCIFAEMVTR	199
Consensus	t h e v v t i l w y r a p e i l l g c k y s t a v d i w s l g c i f a e m v t r	
Homo_sapiens.seq	ALFPFGDSEIDQLFRIFRTLGTPEVSWPGVTSMPDYKPS	239
Gallus_gallus-CDK2.seq	ALFPFGDSEIDQLFRIFRTLGTPEVSWPGVTSMPDYKPS	239
Sceloporus_undulatus_-CDK2.seq	ALFPFGDSEIDQLFRIFRTLGTPEVSWPGVTSMPDYKPS	240
Xenopus_laevis-CDK2.seq	ALFPFGDSEIDQLFRIFRTLGTPEVSWPGVTSMPDYKPS	239
Pimephales_promelas.seq	ALFPFGDSEIDQLFRIFRTLGTPEVSWPGVTSMPDYKPS	239
Ctenopharyngodon_idella.seq	ALFPFGDSEIDQLFRIFRTLGTPEVSWPGVTSMPDYKPS	239
Carassius_auratus.seq	ALFPFGDSEIDQLFRIFRTLGTPEVSWPGVTSMPDYKPS	239
Danio_rerio.seq	ALFPFGDSEIDQLFRIFRTLGTPEVSWPGVTSMPDYKPS	239
Consensus	a l f p f g d s e i d q l f r i f r t l g t p e v s w p g v t s m p d y k p s	
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Gallus_gallus-CDK2.seq	FPKWARGQLSKVVVPLDEIGRLLGCMITYDENKRISAKA	279
Sceloporus_undulatus_-CDK2.seq	FPKWARGQLSKVVVPLDEIGRLLGCMITYDENKRISAKT	280
Xenopus_laevis-CDK2.seq	FPKWARGQLSKVVVPLDEIGRLLGCMITYDENKRISAKV	279
Pimephales_promelas.seq	FPKWARGQLSKVVVPLDEIGRLLGCMITYDENKRISAKN	279
Ctenopharyngodon_idella.seq	FPKWARGQLSKVVVPLDEIGRLLGCMITYDENKRISAKN	279
Carassius_auratus.seq	FPKWARGQLSKVVVPLDEIGRLLGCMITYDENKRISAKN	279
Danio_rerio.seq	FPKWARGQLSKVVVPLDEIGRLLGCMITYDENKRISAKN	279
Consensus	f p k w a r g q l s k v v p l d e i g r l l g c m i t y d e n k r i s a k	
Homo_sapiens.seq	ALVHRRFEDVIMPVVFLR	297
Gallus_gallus-CDK2.seq	ALVHRRFEDVIMPVVFLR	297
Sceloporus_undulatus_-CDK2.seq	ALVHRRFEDVIMPVVFLR	298
Xenopus_laevis-CDK2.seq	ALVHRRFEDVIMPVVFLR	297
Pimephales_promelas.seq	ALVHRRFEDVIMPVVFLR	297
Ctenopharyngodon_idella.seq	ALVHRRFEDVIMPVVFLR	297
Carassius_auratus.seq	ALVHRRFEDVIMPVVFLR	297
Danio_rerio.seq	ALVHRRFEDVIMPVVFLR	297
Consensus	a l v h r r f e d v i m p v v f l r	

(2) *The study heavily relies on specific fish models, which may limit the generalizability of the findings across different species.*

Thank you for your comment. First, we compared the amino acid sequences of CDK2 proteins from fish and other vertebrates, which show over 90% similarity. Moreover, the small size, low cost, and external development of zebrafish make it an excellent model for vertebrate developmental biology. It has been reported that due to the high genomic and molecular similarities between zebrafish and other vertebrates, including humans, many significant discoveries in zebrafish development are relevant to humans (ref. 2). Our study concentrated on CDK2 in zebrafish, and the findings should be valuable for other vertebrates.

Reference:

(2) Veldman MB & Lin S (2008) Zebrafish as a Developmental Model Organism for Pediatric Research. *Pediatr Res* 64(5):470-476.

Recommendations for the authors:

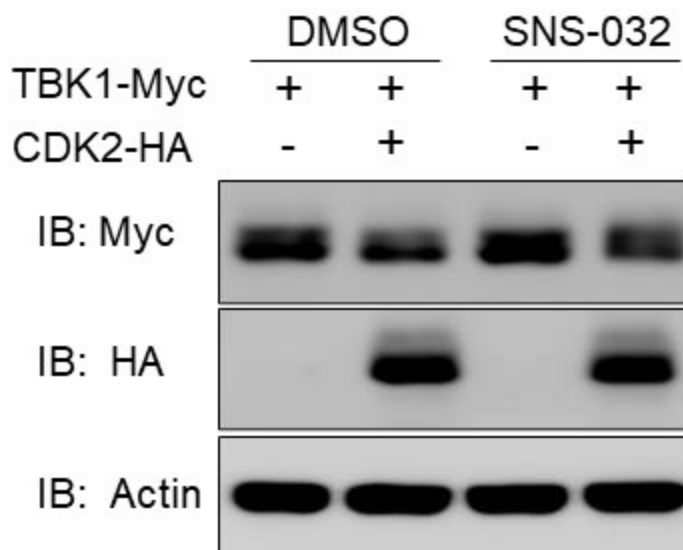
Reviewer #1 (Recommendations For The Authors):

The following additional data/discussion could improve the manuscript.

(1) Investigate whether the catalytic activity of CDK2 is required to regulate TBK1 abundance. It is common for E3 ligases to be directed towards phosphorylated substrates, so it would be of interest to know if CDK2 phosphorylates TBK1 to facilitate its recognition for ubiquitinylation.

We examined the effect of CDK2 on the TBK1 protein after inhibiting its kinase activity with SNS-032 treatment and found that it could still affect TBK1 expression, as shown in the results below (Figure R4). Our previous experiments investigating the effect of CDK2 on TBK1 did not show that CDK2 caused the migration of TBK1 bands (typically, proteins that undergo phosphorylation exhibit band migration). Furthermore, in this study, CDK2 did not function as an E3 ligase; instead, it recruited the E3 ligase Dtx4 to ubiquitinate TBK1.

Author response image 4.



(2) Investigate how CDK2 abundance is regulated by viral infection and whether viral infection impacts cell cycle progression in a CDK2-dependent manner.

In fact, as illustrated in Figure 1, we investigated the changes in CDK2 at both the mRNA and protein levels following viral infection. Our findings revealed that SVCV infection resulted in an increase in CDK2 mRNA and protein expression. Additionally, our earlier reports have indicated that SVCV infection can induce alterations in the cell cycle, resulting in a notable increase in the S phase (Figure 1 of ref. 1). However, whether SVCV infection impacts cell cycle progression in a CDK2-dependent manner will be explored in our upcoming study.

Reference:

(1) Li S, et al. Spring viraemia of carp virus modulates p53 expression using two distinct mechanisms. *PLoS Pathog* 15, e1007695 (2019).

(3) Provide data/discussion concerning the role of fish CDK2 in the regulation of cell cycle progression and whether this process is impacted by viral infection (part 1). Are TBK1 abundance and interferon production differentially regulated across the cell cycle due to the action of CDK2 (part 2).

Thank you for your advice. This concern is addressed in two parts, as follows:

For part 1: To date, there has been limited research conducted on fish CDK2 in the regulation of cell cycle progression. The details are as follows: It has been reported that the kinase activity of goldfish CDK2 significantly increases during oocyte maturation (ref. 1). Furthermore, UHRF1 phosphorylation by cyclin A2/CDK2 is crucial for zebrafish embryogenesis (ref. 2). Additionally, a novel CDK2 homolog has been identified in Japanese lamprey, which plays a crucial role in apoptosis (ref. 3). Red grouper nervous necrosis virus (RGNNV) infection activates the p53 pathway, leading to the upregulation of p21 and downregulation of cyclin E and CDK2, which forces infected cells to remain in the G1/S replicative phase (ref. 4). All this evidence suggests that fish CDK2 plays a vital role in cell cycle regulation, and this process is also impacted by viral infection. Relevant content has been added to the Discussion section in the revised manuscript (lines 389-398).

References:

(1) Hirai T, et al. (1992) Isolation and Characterization of Goldfish Cdk2, a Cognate Variant of the Cell-Cycle Regulator Cdc2. *Developmental biology* 152(1):113-120.

(2) Chu J, et al. (2012) UHRF1 phosphorylation by cyclin A2/cyclin-dependent kinase 2 is required for zebrafish embryogenesis. *Molecular biology of the cell* 23(1):5970.

(3) Xu Y, Tian Y, Zhao H, Zheng N, Ren KX, Li QW. A novel CDK-2 homolog identified in lamprey, with roles in apoptosis. *Fish Physiol Biochem* 47, 189-189 (2021).

(4) Mai WJ, Liu HX, Chen HQ, Zhou YJ, & Chen Y (2018) RGNNV-induced cell cycle arrest at G1/S phase enhanced viral replication via p53-dependent pathway in GS cells. *Virus Res* 256:142-152.

For part 2: TBK1 plays a crucial role in regulating IFN production. Variations in CDK2 activity during different phases of the cell cycle may lead to changes in the expression and function of TBK1. Our findings suggest that heightened CDK2 activity may suppress TBK1 expression, thereby hindering the cell's capacity to produce IFN. Conversely, during the late phase of the cell cycle or in an inhibited state, TBK1 expression may rise, enhancing IFN synthesis and release. In summary, CDK2 is involved in intracellular signaling by modulating TBK1 levels and IFN production, affecting the cellular immune response and cycle regulation—two

processes that are notably distinct at various stages of the cell cycle. Relevant content has been added to the Discussion section in the revised manuscript (lines 377-384).

Minor suggestions:

(1) The authors introduce their study with the consideration that knowledge of fish signaling pathways can inform mammalian biology because mammals evolved from fish. This is not strictly true, since mammals and fish both evolved from an ancient common ancestor and the diversification of signaling in each species likely occurred in response to distinct evolutionary selective pressures.

Thank you for your suggestion. We have revised the statement in the manuscript to eliminate the notion that mammals evolved from fish (lines 98-99). The immune systems of higher vertebrates (e.g., humans) and lower vertebrates (e.g., fish) generally exhibit some consistency, although there are notable differences.

(2) On line 210 and line 276, the authors appear to have misstated the data. CDK2 knockout increases not decreases TBK1 and Dtx4 knockdown abrogated rather than restored CDK2 suppression of TBK1.

Thanks for your reminder, I jumped to the wrong conclusions in these two places (line 204 and line 267) and have changed them as you suggested.

Reviewer #2 (Recommendations For The Authors):

The manuscript has some shortcomings that, if addressed, could improve the overall quality of the article.

(1) Line 63-72, line 77-79, line 88-90- please add additional references for these sentences.

Thanks to your comment, we have added references for these sentences (Line 63-72, line 77-79, line 88-90).

(2) It is of the utmost importance to quantify the data presented in Figures 4J and 5D, as this will facilitate the visualization of the immunoblot.

Thank you for your comment. We have quantified the data presented in Figures 4J and 5D to enhance the clarity of the immunoblot.

(3) The scale in Figure 4E is difficult to discern.

Thanks for your comment. To improve the visual clarity of the image, we have enlarged the scale label in Figure 4E.

(4) In Figure 3B, shCDK2 is shown in italics, preferably in line with other standards such as Figures 3C and 3F.

Thank you for your comment. We have revised the shCDK2 in Figure 3B.

(5) The functions of CDK family members in immunity are hoped to be discussed.

Thanks for your suggestion. We have discussed the functions of CDK family members in immunity (lines 363-387). The details are as follows: Recent studies have demonstrated that CDK activity is crucial for virus-induced innate immune responses. Reports indicate that

CDKs are involved in the Toll-like receptor (TLR) signaling pathway, the nuclear factor- κ B (NF- κ B) signaling pathway, and the JAK-STAT signaling pathway. For instance, CDK8 and/or CDK19 enhanced the transcription of inflammatory genes, such as IL-8 and IL-10, in cells following TLR9 stimulation. CDKs and NF- κ B establish a remarkable paradigm where CDKs can act directly on substrate proteins rather than depending solely on transcriptional control. It has been reported that CDK1 serves as a positive regulator of the IFN-I signaling pathway, facilitating STAT1 phosphorylation, which subsequently boosts the expression of ISGs. Furthermore, inhibiting CDK activity has been shown to obstruct STAT phosphorylation, proinflammatory gene activation, and ISG mRNA induction in response to SeV infection. It is important to note that no evidence suggests the involvement of CDKs in RLR signaling pathways. This study has shown that fish CDK2 functions as a negative regulator of the key kinase TBK1, which is involved in the RLR signaling pathway. A better understanding of the relationship between CDK2 and RLR signaling pathways will enhance our grasp of the regulatory mechanisms of CDKs in antiviral innate immunity.

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